Contents

1 Introduction **3**
- 1.1 Reading suggestions . . . 3
- 1.2 Legend and conventions . . . 3

2 Targets distributions: Am theory **4**
- 2.1 Attacking matrix . . . 4
- 2.2 Slot renumbering . . . 8
- 2.3 Non-standard renumbering . . . 10
- 2.4 Vector notation for Am . . . 11
- 2.5 An example of Am theory application: rocket ships vs steam rams 12

3 How to do calculation a posteriori on a detailed report **15**
- 3.1 Integer damage calculation . . . 15
- 3.2 From integer to percentage . . . 16
  - 3.2.1 Percentage approximation . . . 16
- 3.3 From percentage to integer . . . 17
- 3.4 Purpose of this section . . . 17

4 Damage redistribution **17**
- 4.1 Introduction . . . 17
  - 4.1.1 Moral introduction . . . 17
  - 4.1.2 Introduction . . . 17
- 4.2 A simple test . . . 18
- 4.3 Reserve-Redistribution (RR) . . . 18
  - 4.3.1 Formula for RR . . . 20
- 4.4 Other type of redistribution: nomenclature . . . 21
- 4.5 Slot- Redistribution (SR) . . . 21
- 4.6 Immediate-Redistribution (IR): a rockets vs steam rams explanation . . . 25

5 Kill types **26**
- 5.1 The Example . . . 27
  - 5.1.1 Example of P-kill . . . 28

6 Introduction to open problems: can we predict which kill type will be used? **28**

7 No hephe carabineers vs hephe holpites (open problem) **29**

8 Balloon carriers vs paddle speedboats (open problem) **29**
- 8.1 How to find redistribution patterns . . . 30
- 8.2 A battle vs barbarians . . . 33

9 A rocket ship in the smallest battlefield (open problem) **34**

10 A swordsmen vs barbarian knifemen study (open problem) 35
10.1 Tests 35
10.2 Notation 35
10.3 Tests with unique wave 36
10.4 Explanations and observations 37
10.5 Tests with more waves 37
10.6 Observations 39
10.7 An interesting round 39

11 Columns role in Am (open problem) 40
11.1 Tests with spearmen 42
11.2 Paddle speedboats vs paddle speedboats 44

12 Beyond Am theory (very open problem) 44
12.1 Lunatic first line 44

13 Am theory lacks (very open problem) 46
13.1 The power of an absent second line (very open problem) 49

14 Tests about gyrocopters (very open problems) 50

15 Symmetry issues in gyrocopters: do all units always fire? (very open problem) 54

16 Two round of swordsmen without doctors (very open problem) 56

17 Duration tests (very open problem) 57

18 What about accuracy? 60

19 Other questions 60

20 Other useful facts in tests (bonus section) 60

21 Authors and thanks 61

Ikariam Battle System

AT Barbarossa, Xanto

July 16, 2020

Before read this paper please read the Introduction

1 Introduction

This paper is not a guide. This is a pseudoscientific paper on Ikariam battle system; result of years of playing, testing and reasoning. For battle system I mean the algorithm that calculates losses and left damage on units in a round. There are two main argument: Am theory and redistribution. It can be possible you already know these argument but you address them with an other name.
I try to be as clear as possible in writing this paper but obviously there could be spelling errors, English errors and not perfect explanations.
Moreover this is not a definitive work. Changes , improvements and new sections can maybe added in the future.

1.1 Reading suggestions

If you are terrorized about basic maths, I advice you not read this paper. If you are interested only in making gap against your opponents I advice you not read this paper.
You do not need to read all this paper. I can divide sections according to difficulty:
-First level: sections 1; 2.1; 2.2
-Section level: from section 3 to section 4.3.1
-Third level: section 2.3; 2.4; 2.5; from section 4.4 to section 5.1.1
-Forth level : from section 6

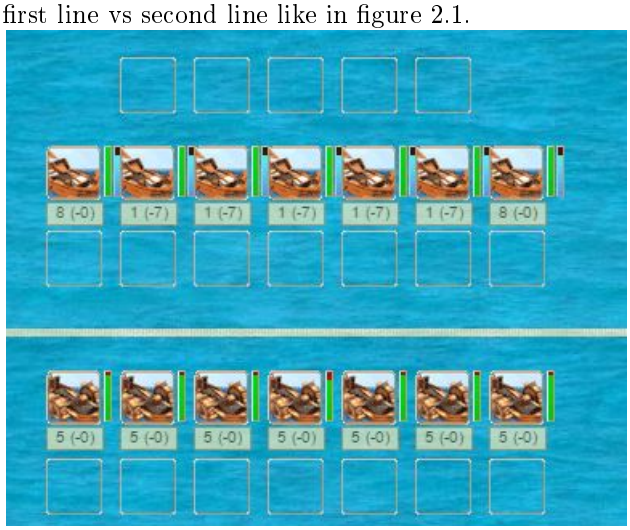
1.2 Legend and conventions

- We will use $\uparrow$ to talk about defender's units shooting attacker's units.
- We will use $\downarrow$ to talk about attacker's units shooting defender's units.
- **Standard renumbering**: we renumber slots according to their "damage position" that means slot 1 is the first slot to be hit and is the central one; slot 2 is on the left of slot 1, slot 3 in on the right of slot 1 and so on this way $[6, 4, 2, 1, 3, 5, 7]$. This is all you need for section 2.1 but in section 2.2 and 2.3 we will talk more about slots renumbering.

2 Targets distributions: Am theory

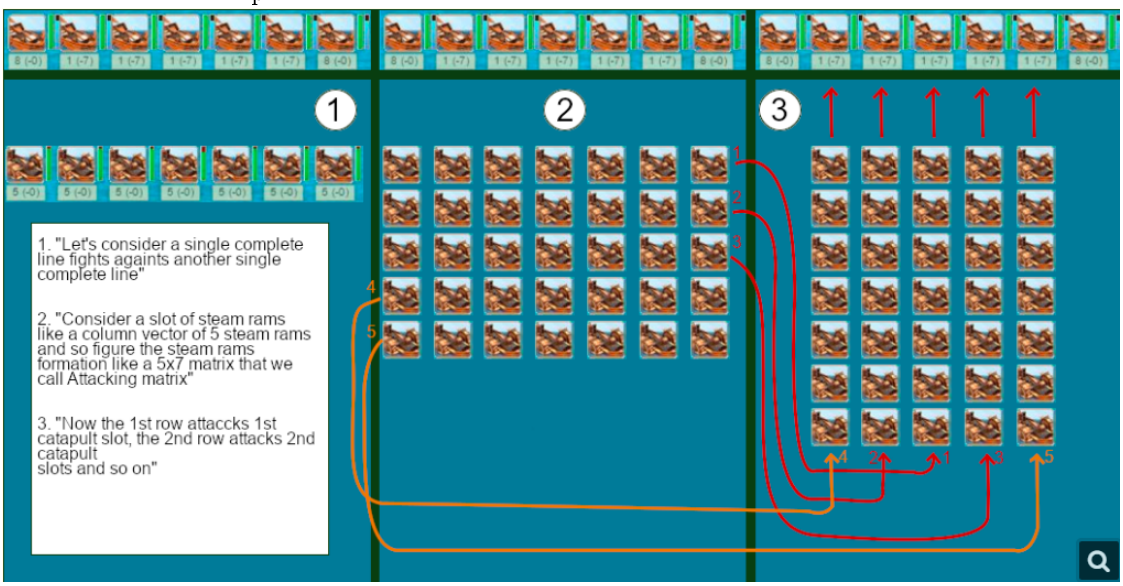
2.1 Attacking matrix

Let consider a single complete line fights against another single complete line, for example first line vs second line like in figure 2.1.



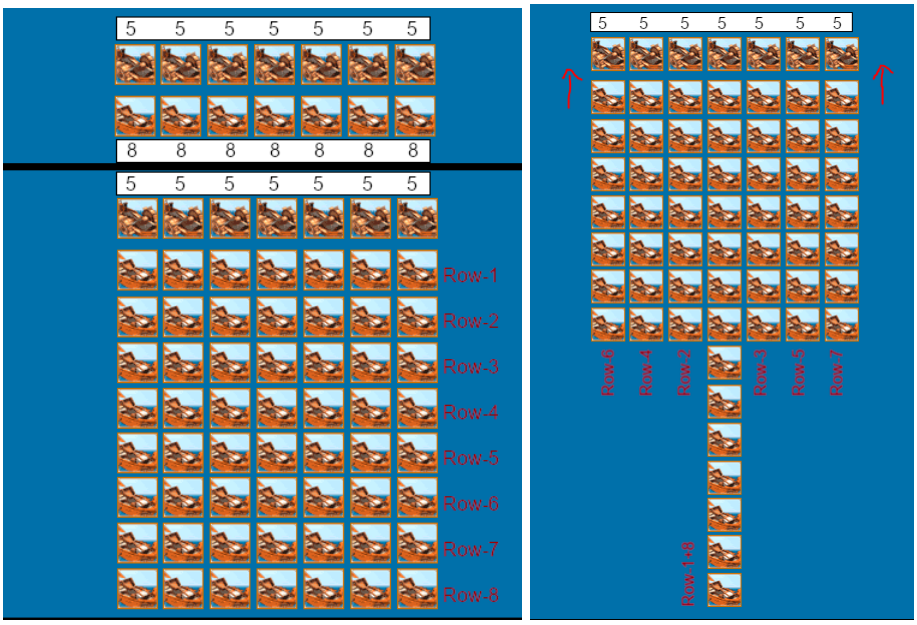
One steam ram kills one catapult ship so 35 steam rams destroy 35 catapult ships, but why external slots of catapults are not hit?

Consider a slot of steam rams like a column vector of 5 steam rams and so figure the steam rams formation like a 5x7 matrix that we call Attacking matrix.[1] Now the 1st row attacks 1st catapult slot, the 2nd row attacks 2nd catapult slots and so on. There are only 5 rows in the Attacking matrix so 6th and 7th catapult slots are not hit.



[1] A matrix is a rectangular array arranged in rows and columns.

Now consider catapults firing against steam rams. In this case the Attacking matrix is a 8x7 matrix (figure on the left) and in fact we observe that all steam rams slots have some damage. Moreover damage of the central slot of steam rams is the highest and the others have all the same amount of damage: this because 8th catapult attacking matrix's row fires on the central slot (figure on the right).



Generally we have a n x m Attacking matrix with $m \leq 7$ and where n is the maximum unit number of slots of a line; for example a first line composed with 6 steam rams slots and a 11- fire ships slot gives the following 11 x 7 attacking matrix.

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

At this point we ignore columns position : in our theory we can do any permutation of attacking matrix columns and final result will be the same.
Furthermore since 50 is the maximum number of units of a slot we can always think a 50 x m matrix with necessaries final zeros rows.

Now if the Attacking matrix hit a line with q slots, the row i shoots on the

slot j so that $j \equiv i \quad (q)$.[2] In a more simple way you can do the integer division between i and q and the rest is j . But in both way you have to pay attention when j is 0 then i shoots on the qth slot: in fact the 7th catapult ships row of the first example shoots on the 7th steam rams slot.

Now we include some screen where you can figure out how this "Attacking matrix theory" works. We will abbreviate attacking matrix with Am.

Observe the ↓ 6x6 rams (flanks) Am against mortar ships (7th mortar ships slot is not hit)

First observe ↓ 4x2 balloons Am against 3rd line. Then observe ↑ 6x5 3rd line

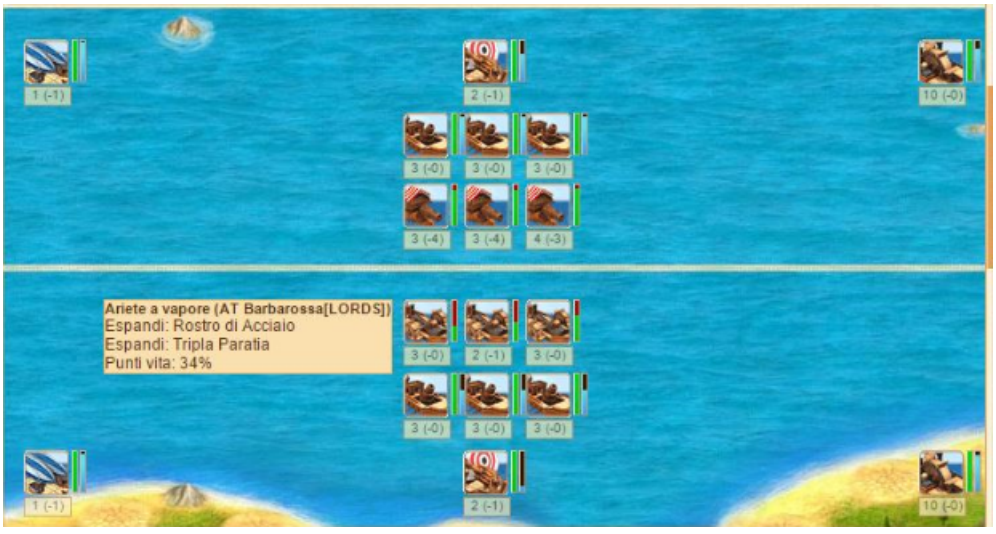
[2] The operation $\equiv$ is the math congruence modulo q (https://en.wikipedia.org/wiki/Congruence_relation)

Am against steam giants

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Steam giants slots 1,2,3 and 4 are hit by 4 third line units while slots 5 and 6 are hit by only one catapult and slot 7 is not hit. This can explain a principal difference of giants deaths ignoring the rest of the round.
In this screen we can also notice balloons vs third line does not follow "standard renumbering". For more details see subsection 2.2.

Observe different ↑ and ↓ balloon carriers Ams producing different 3rd line deaths. Then observe ↓ 3x5 roket ships Am on steam rams: 5 rocket ships shoot the central steam rams slot, 3 rocket ships shoot on the adjacent two steam rams slots and no rocket ships shoots the other steam rams slots.



Observe ↑ rocket ships destroy one fire ship in the central and in the right slot. Assume that ↑ mortar ships do not destroy any fire ship.[3]. Then consider ↓ 7x3 fire ships Am on steam rams: only the central steam rams slot is hit by 7 fire ships (3+3+1 that means line 1,4,7 of the Am) while the other slots are hit by 6 fire ships (3+3, respectively line 2,5 and 3,6 of the Am). This can explain why defender loses only a steam ram in the central slot.

Obiously a wall produce a kill, then the action of the ↑ 3rd and 2nd line Am explains the attacker's 1st line losses . Note that we can ignore how sulphur carabineers works in details. We know that 1 ram kills 1 hoplite but 1 carabineer do not kill 1 hoplite, so we can enter in the very difficult units kill and damage distribution problem but it is not important now. In fact we have 5 hoplites killed in each slot hit by rams and one killed by stone wall, except in slot 1 and 2, then is reasonable that 14 carabineers kill 7 hoplites and 7 carabineers kill 4 hoplites. For this "reasonable" see section 5 (D-kill).

2.2 Slot renumbering

In section 1 we define standard renumbering, as [6,4,2,1,3,5,7]. Unfortunately this is not always true, but renumbering is important in Am theory because the slot renumbered as i is hit by Am's rows whose index have rest i in the division $\frac{\text{index}}{\text{number of slots}}$ and so on. Now, we do not know the combat algorithm, so we renumber a posteriori (=after have seen a detailed combat report), and the idea is we renumber slot 1 the slot that take the most damage, and so on. Now look at the following screen:

[3] Actually we do not have a proof of this fact but we are sure about this as a redistribution fact (see section 4)

First of all observe the attacker's third line and it is obvious its renumbering is [3,1,2]. renumbering for third line, even in land battles is [5,3,1,2,4] with odd and even numbers swapped respect standard renumbering. You can see an example of a land battle in a screen in section 2.1.

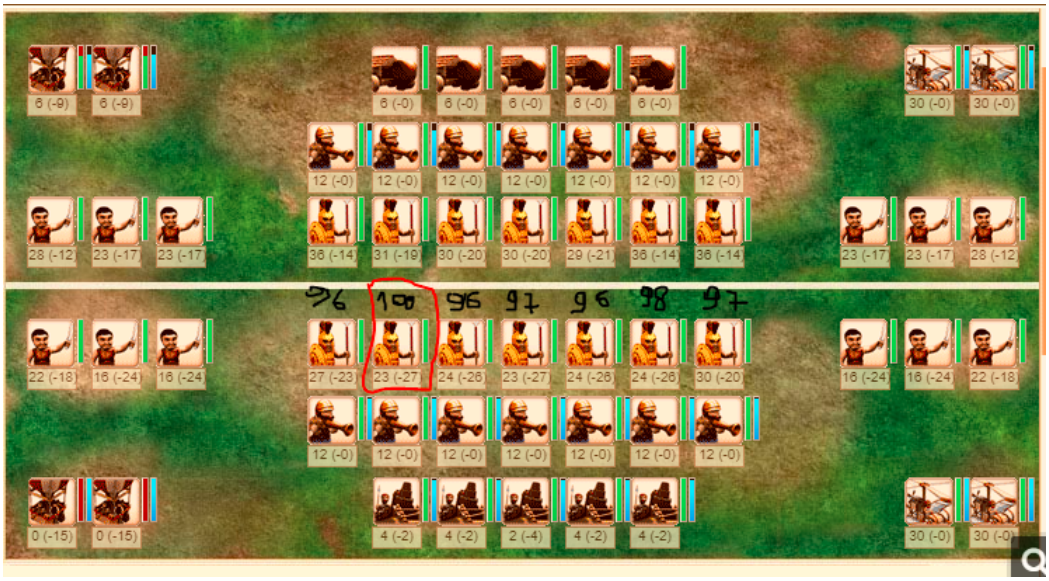
Then we note deployment in the defender third line: the defender has only 5 rocket ships and they are deployed in slots in the same order of " the standard renumbering for that line ". Now, if you play Ikariam you can look at battle field preview, down in the window when you are selecting units for a mission: by selecting units little by little you can see how they deploy in the the battle field. So you see first line and second line deploy [6,4,2,1,3,5,7] while third line deploy [5,3,1,2,4]. Flanks deploy [6,4,2; 1,3,5], balloons [1,2] and anti-balloon units [2,1]. These are the standard renumbering for each line and from now we refer at them with the words standard renumbering.

So standard renumbering is the same that a standard deployment but int the above screen we see deaths on the defender's third line only in slot 1. This is a particular case because slot 2 is incomplete (only 2 rocket ships) and so Am theory described until now is not enough to a correct explanation.

2.3 Non-standard renumbering

In this section we will see 2 examples of a non standard renumbering

Same attacker deployment. On the right screen at round 10 we have standard renumbering, while on the left screen we have 3 steam rams killed on the right of fire ships instead of 2. Also 9 losses for the central fire ships slot are less then we expect. As for percentages at round 10 we have [75,60,54,54,45,60,91] while at round 9 [75,60,54,74(fire ships),54,60,91]. We can explain this situation with a non-standard renumbering for round 9: $[6, 4, x, 3, z, 5, 7]$ with $z \in \{1, 2\}$ $x \in \{1, 2\}$



Some following sections are necessary to undersand this second example.
$\downarrow$ rams kill $[5,5,5,5,5,5,0]$ hoplites, then $\downarrow$ carabineers kill $[4,7,7,7,7,7,4]$ hoplites.
$\uparrow$ mortars kill $[0,4,5,5,4,0,0]$ hoplites, then $\uparrow$ carabineers kill $[0,6,6,6,6,6,0]$ hoplites. You can control this assumption with simple tests; in particular for carabineer we have screens in section 5, and hoplites losses do not change if hoplites are without hephaistos.
So after second line's fire we have [50,40,39,39,40,44,50] for the attacker and [41,38,38,38,38,38,46] for the defender. According to Am theory, the vector notation for $\downarrow$ Am of hoplites is [40,44,45,47,44,42,40]. But this vector does not correspond to the symmetry in the screen. So for in the defender's first line we have a non standard renumbering $[6,2,x,1,y,5,7]$ where $\{x,y\}=\{3,4\}$
According to Am theory the vector notation for $\uparrow$ hoplites is [37,38,43,43,43,37,36],
In this case renumbering is standard and the difference of losses can be explained only at a higher level (probably section 11 and 12).

2.4 Vector notation for Am

In order to avoid writing big matrices, often in the continuation of this paper we will use a vector notation for Attacking matrices **with attention to targets position** namely with the **Standard renumbering** for the component of a vector $v=[v_6,v_4,v_2,v_1,v_3,v_5,v_7]$
In a general way we can image a n x m matrix and compute an horizontal vector A with length equal to Am's columns.

$$A_i=\sum_{k,j}(Am)_{k,j} \qquad k\equiv i(m)$$

namely the sum of elements of the rows with index whose rest in the division for m is i.[4]
As example we write the 12 x 7 Am and the corresponding horizontal vector for a line of 84 sulphur carabineers.

$$Am = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \qquad A = \begin{bmatrix} 7 & 14 & 14 & 14 & 14 & 14 & 7 \end{bmatrix}$$

in fact $A_1 = A_2 = A_3 = A_4 = A_5 = (1+1+1+1+1+1+1) + (1+1+1+1+1+1+1) = 7+7 = 14$ and $A_6 = A_7 = (1+1+1+1+1+1+1) = 7$

The meaning of vector A is the following : the slot i of the first line of the opposite formation is hit by i sulphur carabineers.
As other example you can go back to schemes with steam rams and catapult ships in section 2.1: the vector notation for steam rams Am is $[0, 7, 7, 7, 7, 7, 0]$ (sum of columns of the rightest figure) while the vector notation for catapult ships is $[7, 7, 7, 14, 7, 7, 7]$ (sum of columns of the rightest figure).

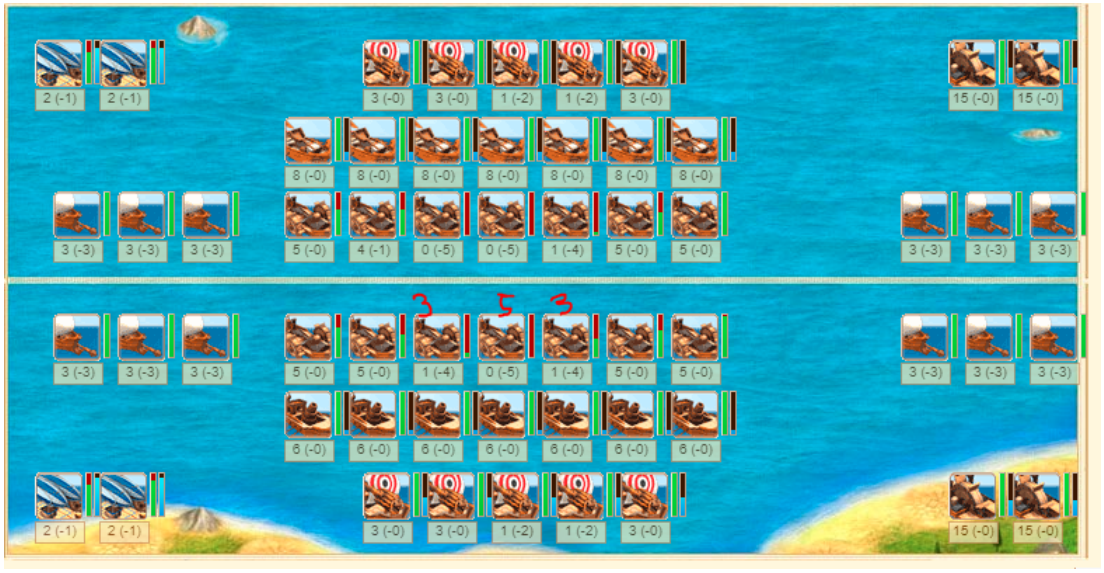
2.5 An example of Am theory application: rocket ships vs steam rams

In this section thanks to Am theory we will prove the following statements: with hephaistos **a row of**

- 5 rocket ships kill 3 steam rams
- 4 rocket ships kill 2 steam rams
- 3 rocket ships kill 1 steam ram
- 2 rocket ships kill no steam ram

So let see the following 3 screens. According to Am theory, on each screen we write in red the number of $\downarrow$ rocket ships that fire in each steam rams' slot. We obviously do not write anything for slots not hit by roket ships.
At this point we also assume that second line (both mortar than catapult ships) do not kill any steam rams.

[4] $\sum$ is a summation operator.



Let's start with this screen and observe the attacking steam rams deaths. We observe 5 deaths in slot 2 but only 4 deaths in slot 3 and both slots are hit symmetrically by $\uparrow 3$ rocket ships. Because mortar ships do not cause any death we can explain this difference only by $\uparrow$ steam rams. That means attacker's slot 3 is hit by at least one less $\uparrow$ steam ram than slot 2. Then repeat the same argument observing attacker's steam rams slots 4 and 5.

Now we will write a vector with the number of $\uparrow$ steam rams fire on each of five attacker's slots $\quad v = [v_4, \quad v_2, \quad v_1, \quad v_3, \quad v_5]$

Let consider a defender's steam rams slot S: according to Am theory S can hit up to the attacker's slot number 5. But according to priority targets order the 5th attacker's slot is the last to be hit, so if it is hit then all other attacker's slots from 1 to 4 are hit by S. So $\forall i \quad v_i \geq v_{i+1}$ Then because the defender have four slots of 5 steam rams alive at the end of the round, $\forall i \quad v_i \geq 4$. Moreover according to Am theory $\forall i \quad v_i \leq 7$

Because our first observation we made $v_4 \geq v_5 + 1 \quad v_2 \geq v_3 + 1$ and follows that $v_1 \geq v_5 + 2 \quad$ so $v_5 \leq 5 \quad$ (if $v_5 \geq 6$ then $v_1 \geq 8$ contraddiction).

Now suppose $v_5 = 5$, then $v(1) = [6, 7, 7, 6, 5]$ and the corrisponding Am is

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

However this Am is wrong because columns 2 and 3 (remember standard target renumbering) are not symmetrical but we know in both slots fired 3 rocket ships so these columns must be symmetrical. Remember this is the $\uparrow$ steam rams Am and it contains steam rams still alive after $\downarrow$ third line and second line fire.

So if x=4 then we can have three possible v : $v(2) = [5, 7, 7, 6, 4]$ and $v(3) = [6, 7, 7, 6, 4]$ and $v(4) = [5, 6, 6, 5, 4]$ If you write the corresponding Am for v(4) you notice that in defender slot 1 there are 0 alive steam rams but this is not possible in fact, even with hephaistos, the total damage of 5 rocket ships plus 14 catapult ships is less than the total life of 5 steam rams (2713<2880) and so at least one steam ram must remain alive. Note that the inequality remain true

with 7 mortar ships instead of 8 catapult ships (2657<2880). Exercise: why I write 14 catapult ships?
Moreover we can exclude $v(2)$ with a symmetry argument like $v(1)$, so the only correct v is $v(3)$ and this is the corresponding Am:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Observation: we could also exclude $v(4)$ with a symmetry argument like $v(1)$. Now the columns of the Am give the number of steam rams killed by rocket ships: a 0 in the Am correspond to a steam ram killed.
I also include this bonus screen as a direct proof of our statements:

The second screen:

It is the same than the first screen except for the 5th attacker rocket ships slots and so for the number of rocket ships firing on the defender's slot 3 : only 2 rocket ships. Consequently this causes the different number of deaths in the defender's slot 3: one less deaths. That means two rocket ships kill 1-1=0 steam rams.

Now the third screen:

Attacker's steam rams deaths are more than in the first screen, but the defender's third and second line is exactly the same: so in order to have more damage we need more ↑ steam rams firing, so more defender's steam rams alive after ↓ third and second line fire. So think about the vector v after the first screen: in this case we call it $v(5)$ and we have:
$v(5)_3 > v(3)_3$ and so $v(5)_3 = 7$. This is enough to say that 4 rocket ships kill 2 steam rams : write the corresponding Am accoring to statement previously proved and with the first 3 row with 7 ones.

3 How to do calculation a posteriori on a detailed report

"A posteriori" means we first look at the detailed report with losses and life percentages on slots and then we calculate what happened or in other words we find some calculation fitting with the results we see in the battle report. On the contrary "a priori" means we calculate the result of a battle before seeing the report trying to predict the result. In this section we will talk about approximation and percentages issues.

3.1 Integer damage calculation

Let A the attack of the attacker unit; u the up of the attack; a the armor of the target; v the up of the armor of the target; then the damage d

- without Hephaistos $d = (A + u) - (a + v)$
- with lv 5 Hephaistos $d = [(A + u) * 1,2] - (a + v + 2)$ and then you have to take an **appropriate approximation**

These formulas are not complete, in fact you can consider different hephaistos levels and maybe the attacker can have hephaistos but the defender not. In the formula the influence of hephaistos on the attacker is $*1,2$ while in the defender is $+2$ so you can easy adjust the formula according your particular case.
Examplese: a sulphur carabineer inflicts (29+3) - (1+3)= 28 damage on an hoplite; (29+3) - (1+3+2) = 26 damage on an hephaistos hoplite; a sulphur carabineer with hephaistos inflicts 32 damage on an hephaistos hoplite (32,4 $\longrightarrow$ 32); a paddle speedboat with hephaistos inflicts 14 damage on a balloon carrier (((12+6)*1,2) - 8) = 13,6 $\longrightarrow$ 14
WARNING: what does it mean " appropriate approximation" ? The approximation was not tested on all Ikariam units so it is possible to be different in some case. In fact in the case paddle speedboat vs balloon carriers we have an upper approximation from 13,6 to 14 but in the case of swordsman vs barbarian knifemen we have a lower approximation from 15,6 to 15 even if 16 is nearest than 15 to 15,6. So an appropriate approximation could be done only a posteriori after some tests and at this point we can not generalize.

3.2 From integer to percentage

In a detailed battle report we can see life percentage on each slot. That percentage of a slot is made considering the total life of units alive in that slot. For example suppose a slot of 2 balloon carriers with 0 deaths and 140 damage. The total life of balloon carriers is 280 so life percentage is 50% (and it is equal to damage percentage). Now suppose a slot of 3 balloon carriers with 1 death and 84 damage. The total life of alive units is still 280 but the real remaining life is 280 - 84 = 196 corresponding to 70% life. Examples with balloon carriers are easy because with hephaistos a paddle speedboat does as damage exactly $\frac{1}{10}$ of a balloon carrier's life; but in general you can use the proportion
life percentage : 100 = real life : total life

3.2.1 Percentage approximation

Observing detailed battle report we are usually interested in damage percentage and it is 100 - life percentage. Anyway life percentages on reports are always integer and that means there is an approximation on percentages. After a lot of calculations we can assert : **damage percentage is approximated on the upper integer**, and consequently life percentage is approximated on the lower integer. For example 1 damage on a total life of 407 means 0,24% and it is approximated on 1% damage and 99% life. A damage > of 99% is approximated on 100% and 0 % life like in the following screen.

3.3 From percentage to integer

In a test usually it is necessary to go back to damage from percentage. Consider the same example of 1% damage on 407 total life. We have seen that 1 damage would cause 1% damage, but also 2 damage would cause 0,49% and so 1% damage and also 3 and 4 damage would correspond to 0,73% and 0,98% damage approximated to 1% damage. So a posteriori we do not know precisely how much damage, but we can produce an interval of damage, in this case [1,4]

3.4 Purpose of this section

Trying simple a priori calculation we noticed we were not able to predict results of a battle. So the only way to find how the battle system works is starting with the approach of section 3.3. Unfortunately in a lot of cases we are not sure about the " appropriate approximation" and percentage gives us a too big interval so that is impossible to continue a battle analysis with more abstract theories; but ignoring these issues you risk to produce completely wrong conclusions.

4 Damage redistribution

4.1 Introduction

4.1.1 Moral introduction

This section talks about a non-intuitive argument: imagine soldiers on a battlefield. At a point one is hit and fall down. Then he calls his fellow soldiers and pass them some of his injury and pain. As a result all soldiers have some injuries but the hit one picks up.

4.1.2 Introduction

First notice what happens to stone walls: in a round some slots are destroyed, others are only damaged. Then at the start on the next round damage is redistributed on all slots of the wall with a simple division (total damage/number of slot not destroyed).[5] This is a general behavior that you can notice in each line if you do not have doctors/tenders in your army/fleet. Moreover in my opinion remembering exactly which unit is damaged and which not is harder and more expensive than doing a simple redistribution (=division) for an computer system; this is a moral reason why in this paper we focus on redistribution.
Anyway the main idea of redistribution is " new damage = total damage/total units".
I use the term "redistribution" like something happens after a "distribution". The idea is the attacker fires and distributes damage how he wants, but then the defender moves some damage on other targets: he redistributes damage.

[5] For a very detailed guide about damage system on the wall you can try to translate this work by Xanto: https://forum.ikariam.gameforge.com/forum/thread/5262-base-guida-calcolare-il-danno-alle-mura/

4.2 A simple test

I do not include a screen of the following test because from a graphic point of view is unhelpful. So a side, suppose the attacker, is without hephaistos while the defender is with hephaistos. The battle use the smallest field of the game (3 first line slots). The attacker has 30 upped steam giants while the defender has 1 upped spearman and some non-upped slingers in order to do more than 1 round. The spearman inflicts 2 damage on a steam giant; slingers do not cause any damage because giants' armor is better than slingers' attack value. The spearman is killed at the first round by giants. At the end of round 1 we have the following life percentages for giants slots: [100%, 99%, 100%]. Then in the second round there are only slingers and giants, so at the end of this round giants have the same total damage of the previous round but life percentages are: [99%,99%,100%]. Now do not panic and remember section 3: damage does not have increased in fact both 2 and 1 damage gives the same damage percentage. But slot 2 now is damaged and it was not previously ! Of course if you think of a battle damage redistribution is totally unrealistic but the battle system have given one damage to the central slot and one other damage to slot two.

4.3 Reserve-Redistribution (RR)

At this point we affirm that redistribution involves units in reserve that means damage is redistributed even with units in reserve. Look at theese following screens.

First of all remember that doctors are not able to handle steam giants. Then we observe that in both screen no steam giants are destroyed: so only 2 situations are possible: giants deployed at round 13 are the same of round 12 or a mysterious permutation of giants between deployed and reserved happens.
Now we observe the life percentages on steam giants (numbers in black): at the end of round 13 they are less damaged then at the end of round 12. How is it possible? Of course steam giants are not able to cure itself but they do redistribution. Damage we see in the first screen is damage that only giants deployed have at the end of round 12. At the beginning of the round 13 a division of the type "total steam giants/total damage" give each (of 277) giant only a few damage. So same damage is stored in reserve giants and only a few is given at the deployed ones. Then obviously in round 13 barbarians do significantly less damage than in round 12 and so the final result.
Now we do not have a formal way to confute the second theory of permutation between reserved and deployed units because the result is the same: some unhurted units are deployed while damaged units gain reserve, barbarians do significantly less damage in round 13 and so the result. Moreover redistribution could happen at a slot level istead at a global level, but again the result would be the same. However it is easy to prove that a complete permutation does not happen: you can send 300 steam giants against a over gl barbarian formation and observe losses and damage. If every deployed giants would be swapped with reserved ones losses and damage on the 1st and 2nd round would be the same, but it is not the same.

Now a clarification on units in reserve. If you look on the left of a detailed combat report, under reserve you can see all units are not deployed in the battlefield. But there there is no distinction between units of different waves.

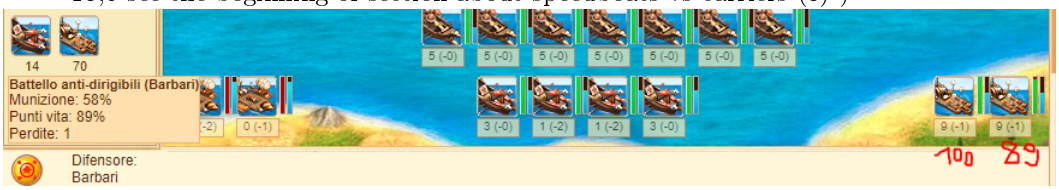
Instead when we talk about redistribution with units in reserve we mean with units in reserve but of the same wave. So if you have a wave of 120 gyrocopters and a wave of 3000 gyrocopters in battle, redistribution is done for these two waves separately; so if the wave of 120 is deployed first (or was already in battle when the second wave of 3000 arrived) the redistribution affect only the remaining gyrocopters of the 120-wave, so at the next round there will be significantly more deaths of gyrocopters even you have more than 3000 in reserve.

4.3.1 Formula for RR

So we now introduce a formula to calculate reserve redistribution
Let D be the total damage of a wave at the end of a round, let N the number of units of the same type of that wave alive, let s the number of units of that wave deployed in a slot at the following round and let d_s the damage of slot with units of that wave at the start of the next round:

$$d_s = \frac{D}{N} * s$$

In this formula, when $d_s < 1$ we have some slots with 1 damage until the total damage is D. Moreover damage is given to slots according to the standard renumbering.
This formula focuses on redistribution on different slots (not on units). We know damage is assigned to units as integer but we do not know how is assigned to units in a situation for example of 5 damage in a slot of 30 units.
As example you can do a 2 rounds battle: first round you send a wave of gyrocopters, second round you send a wave of at least 30 balloon and observe damage on the deployed gyros. That damage is the result of RR because in the second round opponent's gyrocopters fire only against balloons. So the more gyrocopters you send in your wave, the less deployed gyros would be damaged in the second round.
We also include this following example but for a better understanding you need also section 5 :
Paddle speedboats vs barbarian paddle speedboats. Attacker has hephaistos so a speedboat does 19 damage (for the upper appropriate approximation from 18,6 see the beginning of section about speedboats vs carriers (8))



Using section 3 we calculate 89% life in the right barbarian slot corresponds to [19,20] damage interval. (At this point we can suppose 19 that is exactly a shot of a speedboat, but this time using RR both 19 and 20 will produce the same result). So we have [19,20] damage on a total of 88 barbarian speedboats still alive. Using RR formula we calculate damage of a barbarian slot at the start of the next round: $\frac{19}{88} * 10 = 2,159$ $\frac{20}{88} * 10 = 2,273$ Both with 19 or 20 we have a result between the two integer 2 and 3. We do not know if is a general rule doing an upper approximation for damage on slots after RR formula, but this time it fits with the following explanation: after RR formula

suppose an other redistribution in the slot happens and 3 barbarian units obtain 1 damage (similar to SR, see next subsections). Then 10 speedboats fire against a barbarian slot. 3 speedboats hit 3 damaged barbarians and kill them, then 3 couple of speedboats kill a barbarian each without any remaining damage (D-kill, see section 5) and finally the last speedboat fire without kill anyone. The result is 6 barbarian deaths and 19 damage on the slot, corresponding to 24% damage –> 76% life.

Now the total damage on slots is $19 * 2 = 38$ (this time from a posteriori calculation we do not have a damage interval, only 19 is compatible with 24% damage). But in reserve we still have $19 - 6 = 13$ damage, so the total damage on 76 barbarian speedboats is $38 + 13 = 51$. Using RR formula at the start of the next round each barbarian slot have $\frac{51}{88-6*2} * 10 = 6,7 - - - > 7$ damage. So 7 attacker's speedboats kill directly 7 units, then 2 speedboats perform D-kill and finally the last speedboats fires without kill anyone

You can continue calculus by yourself to find the intuitive result that the next round all barbarian speedboats have at least 1 damage and so they all die.

4.4 Other type of redistribution: nomenclature

In this small section i will give a name to different types of redistribution in order to refer to and discuss about them later easy.

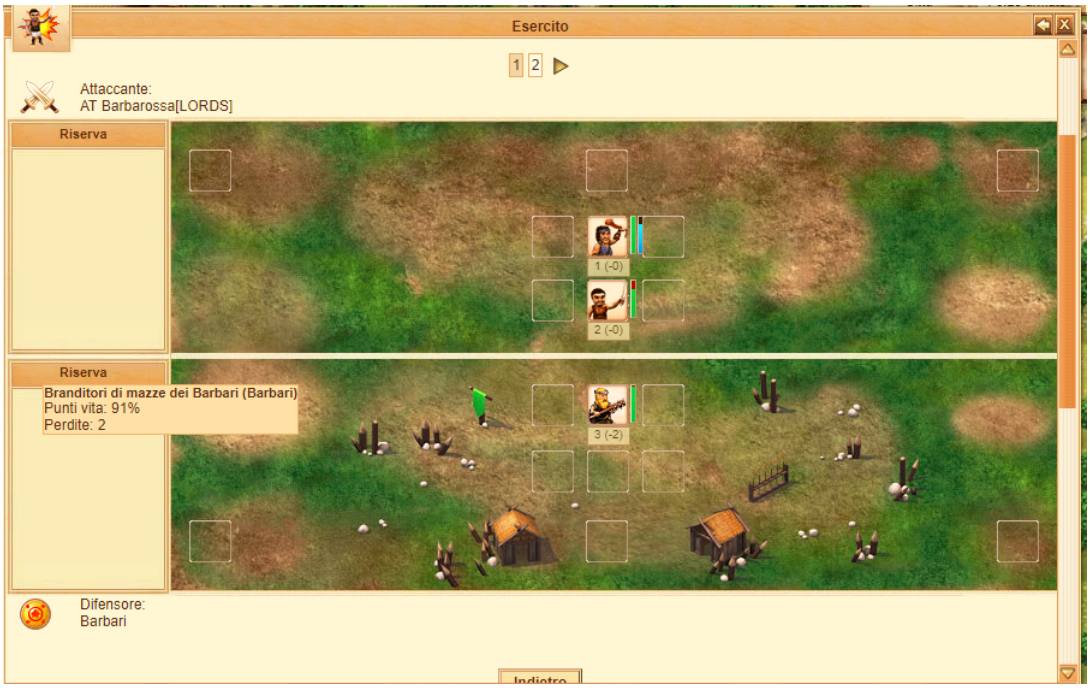
- Reserve-Redistribution (**RR**) : happens between rounds, see previous sections
- Immediate-Redistribution (**IR**) : happens in a slot immediately after a unit fired, even if there are other units of the same line that have to fire in the same slot.
- Slot -Redistribution (**SR**) : happens in a slot after a line fired, before the next line start to fire.

4.5 Slot- Redistribution (SR)

First we will analyze the following screen

A barbarian have 12 life and 1 armor. An upped swordsman have 12+3 attack while an upped slinger have 3+3 attack. Hephaistos is not active. So the slinger makes 5 damage on barbarians, then swordsmen kill one barbarian each. We now write some possible round development.
1) The slinger hit a single barbarian; when swordsmen attack there are one hurt barbarian and four unhurt barbarians. Swordsmen hit the hurt barbarian first and then other 3 barbarians–> the remaining unique barbarian should have 0 damage =100% life but in the screen we see 91% life so this battle development is wrong.
2) The slinger hit a single barbarian; when swordsmen attack there are one hurt barbarian and four unhurt barbarians. swordsmen hit first the unhurt barbarians so the remaining barbarian should have 5 damage of the slinger. But 5 damage on 12 life means 42% damage means 58% life but in the screen we see 91% life so this battle development is wrong.
Now 9% damage means 1 damage and so
3) The slinger hit a single barbarian, then before swordsmen attack redistribution on barbarians happens: 5 damage is redistributed on each barbarian so each barbarian have 1 damage. Now swordsmen attack and kill 4 barbarians. The unique remaining barbarian have 1 damage.
We can prove redistribution with these other screens with different number of swordsmen:

Here we have 4 damage on barbarians (one on each barbarian).



Here we have 3 damage on barbarians (one on each).

And finally here we have 2 damage on each barbarian.

We call this redistribution Slot-Redistribution and we can generalize in this way: after a line fired on another target line, redistribution can happens separately on every slot of the target line. So chronologically : second line fires-> redistribution on first line slots-> first line fires; but also third line fires-redistribution on first line-second line fires; and so on.

Unfortunately SR does not always happen, as for example in the following screen:

In this test barbarians have 1 armor, archer is upped (5+3 damage) so he causes

7 damage on barbarians; spearman is upped (4+3 damage) so he causes 6 damage on barbarians . Observing that 9% damage = 1 damage the only possible battle development is archer and spearman attack the same barbarian (13 damage) so they kill one barbarian and 1 damage "left over". So no redistribution but we can ask ourselves why 1 damage is unrealistically sent to the second barbarian. Unfortunately a whole understanding of battle development even in these simple test is far but with nomenclature we will introduce in section 5 we call this above situation P-kill

We can repeat the test with an hoplite instead of a spearman and in this case redistribution happens:

In fact 15% damage means 3 damage and this fits with a damage redistribution [4,3] and then with a hoplite attacks and kills on the most injured barbarian.

4.6 Immediate-Redistribution (IR): a rockets vs steam rams explanation

In this short paragraph we see how IR can explain the statement we talk about in paragraph 2.5.
A rocket with hephaistos makes 439 damage on a steam ram with hephaistos. A steam ram has 576 life, but we knew 2 rocket ships do not kill any steam ram. So a rocket ship fires, then redistribution happen immediately and we have about 87 damage on each steam ram of the slot. Then the second rocket fires and again we have redistribution and after that each steam ram has about 174 damage. Now the third rocket fires and we note 174 + 439=613 >576 so a steam ram dies. The same happen when the fourth and the fifth rocket fires.
We will see a similar behavior in the paragraph " balloons vs rams".

5 Kill types

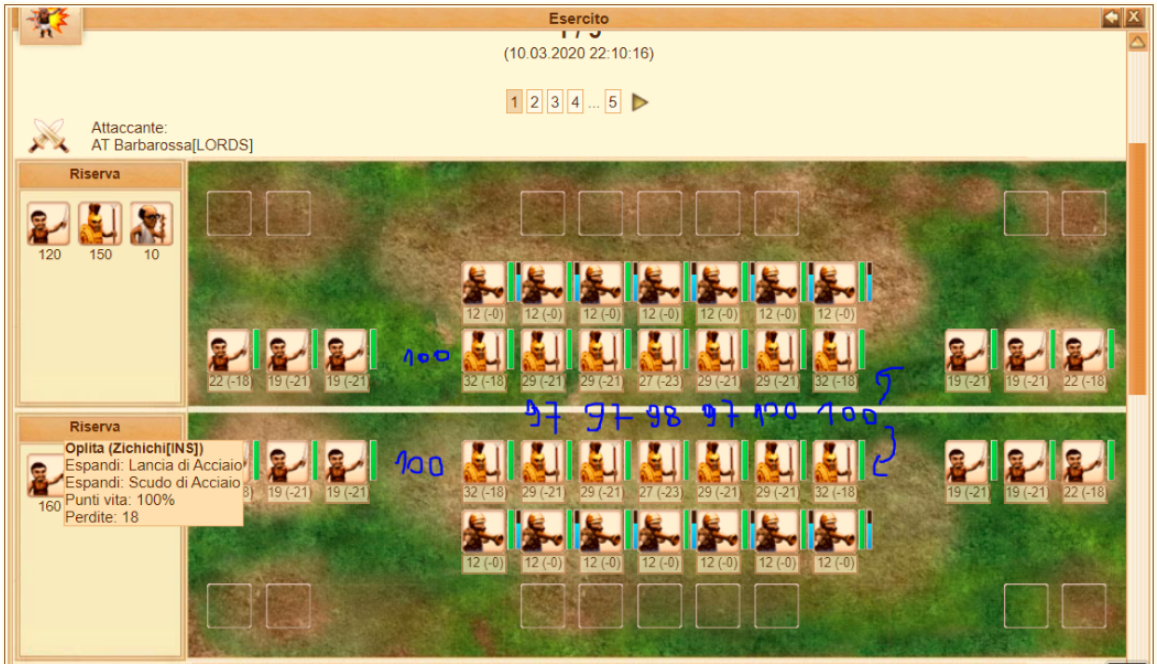
In the Ikariam battles I have found some different types of kill. I use this brief paragraph to give them a name I will use later. WARNING: this names are only conventions that allow us to easy talk in the continuation of this work. They have been created a posteriori after tests and according to new tests and theories they could be modified.

- One-Shot-kill (**1-kill**) : exactly a units kill exactly a target . No damage remains on survivors. Examples: a ram kills a hoplite; a rocket ship kills a fire ship.
- Direct-kill (**D-kill**) : attacking units hit sequentially a target unit until they kill it. No damage remains on survivors. Suppose n attacking units kill 1 target ($n * \text{atk} \geq \text{life}$); then D-kill can happen when the number of attacking units is multiple of n. Example : 14 hephaistosed carabineers kill 7 hoplites (see next section).
- Pig-kill (**P-kill**) : like D-kill but if the sum of attack values is bigger than target's life, the difference is reassigned in some (mysterious) way to survivors. Usually we can see P-kill only when only a few attacking units are involved. (The name Pig-kill derives from a local expression : " Del maiale non si butta via nulla " means no section of pig , like of the damage in this case, is throw away). Example: see section 5.1.1.
- Indirect-kill (**I-kill**) : first the attack value of attacking units is summed and then the number of units killed is calculated by the division : $\frac{\text{sum of attacks}}{\text{life of a target}}$ This way could happen only if the above division is integer. Examples: 7 hephaistosed carabineers vs hoplites (see next section) or for an other example see the section about swordsmen vs knifemen.
- Immediate Redistribution- kill (**IR-kill**) : a kill involving only a line and using IR; some attacking units perform IR until target's life becomes less than attack value and so some other attacking units fire on only one target each and kill it. Example : rocket ships vs steam rams; balloons vs rams (see a following section). We can have a variant : Pig-Immediate Redistribution -kill (**PIR-kill**) when other than IR-kill the unit performing last shot has an atk value bigger than tagert's remaining life and the difference is reassigned in some (mysterious) way to survivors.
- Magical-kill (**M-kill**) : in this case we start like P-kill but then some damage reassigned to survivors is used in some mysterious (or magical way) to cause other losses. Usually we have M-kill when any other kill types explanation fails. Moreover usually some damage is lost: in other words the sum of killed units' life plus remaining damage on the slot is less than the sum of attack values. Examples: gyrocotpers vs gyrocopters; gyrocopters vs balloons; swordsmen vs swordsmen.

We will accurately see examples of D-kill and I-kill in the following section.

5.1 The Example

In this section we will study a unrealistic round, without third line. The situation is perfectly symmetrical, so we do not any reference to attacker or defender. Life percentages of hoplites are in blue on the screen.



Damage by carabineer on hoplite: 32; damage by hoplite on hoplite: 19 (hephaistos lv 5 on each side).
As in section 2.4, the attacking vector for carabineers is C=[7,14,14,14,14,14,7].
Now we use the last example in section 2.1: the screen with walls in front of carabineers. So 14 carabineers kill 7 hoplites and this is an example of D-kill: 2 carabineers attack one hoplite and kill him. And 7 carabineers kill 4 hoplites with a I-kill $(32 * 7) = 224$; $224/56 = 4$
So after second line fire, we have [4,7,7,7,7,7,4] hoplites deaths, and so [46,43,43,43,43,43,46] hoplites still alive. With these hoplites we construct the Am, and the corresponding vector [42,45,45,49,45,42,42].
Now 3 hoplites kill one hoplite (D-kill).If we use a "+" to indicate an hoplite's hit that does not cause any death we have the following vector of deaths: $[14, 14++, 14++, 16+, 14++, 14, 14]$. In conclusion if we sum deaths by second line and deaths by first line we have exactly numbers like in the screen. Anyway we write calculation of percentages:
2 hoplites$\rightarrow$ 38 damage $\rightarrow$ $3800/(56 * 29) = 2,34\% \rightarrow 3\%$ means 97% life;
1 hoplite $\rightarrow$ 19 damage $\rightarrow$ $1900/(56 * 27) = 1,26\% \rightarrow 2\%$ means 98% life.

At this point we could also observe D-kill on the externals slots of swordsmen where 36 kill 18, and r-kill on the other swordsmen slots; but now is more important to comment computation already done. In fact in this example all work perfectly with simple calculations.
When 2 carabineers kill 1 hoplite we have 64 damage but an hoplite has only 56 damage so we have 8 damage left. In the same way when 3 hoplites kill 1

hoplite we have 1 damage left. In this case all damage left from D-kills is lost, but this is not true in general and in fact there is P-kill More precisely we do not have a formal proof that in every kill of this round the left damage is lost, but our previous explanations fits with the screen, even if in the battle algorithm happens something more complicated.

5.1.1 Example of P-kill

Now an example of P-kill where left damage is not lost:

According to Am theory the defender's slot of flanks 6 is hit by only 2 rams: the total attack (calculated with hephaistos) is more than the life value of a ram ($80 * 2 > 154$) but life is 99% corresponding to 6 damage left recounted. But wait, how can be sure 80 is the "appropriate approximation" (see section 3)? An exact count gives us $(75 + 6) * 1,2 - (9 + 6 + 2) = 80,2$ attack value. If we approximate at 81 we have a total damage of 162 and so a damage left of 8, but 8 damage correspond to 1,03% –> 2% (see section 3.2.1) and in the screen we have 99% life so only 1% damage.
Notice this example works because of the row with only 2 rams. In the standard case we have 6 rows with 6 rams each. And each slot of flanks have 3 losses and life percentage of 96% means 4% damage. 4% damage corresponds to [14,18] as damage interval. We see $6 * 3 = 18$ (3 because of 3 killed rams) is included in the interval but we cannot be sure of the real amount of damage.

6 Introduction to open problems: can we predict which kill type will be used?

From this section we will report some battle and round analysis and we will try to explain some situations using Am theory and Redistribution, but they are not sufficient to a whole understanding and explanation of the battle system. In other words the above theory of this paper is not enough but we will show our interpretation of some interesting cases without knowing if it is true. "Open" means without a (certain) solution.

The main question we are not able to answer in general is what type of kill happens in which situation. We will show in the next sections how the kill type can completely be different even for very similar situations.

From this point it is possible that hephaistos and hephaistosed will be abbreviate with hephe.

7 No hephe carabineers vs hephe holpites (open problem)

First look at this screen: ↑ carabineer have no hephaistos while ↓ hoplites have it.

A piece of wall kills one hoplite opposite to it. Then apply Am theory to ↑ third line and find $[0, 4, 5, 5, 4, 0, 0]$ hoplites killed due to third line fire. So subtracting contribution of walls and third line we see ↑ carabineers kill $[0, 6, 6, 6, 6, 6, 0]$ hoplites. It means 14 carabineers kill 6 hoplites. A carabineer does 26 damage on a hopilte ad we have : $(26 * 14)/56 = 6.5$ so we can say M-kill.
Then 7 carabineers kill no hoplites and this can be explained in 2 ways: first way is IR, second way is each carabineer hit a different hoplite.

8 Balloon carriers vs paddle speedboats (open problem)

Paddle speedboat damage on balloon carriers is 14 with hephaistos on each side. We can be sure of this approximation with tests in the smallest naval battlefield: there, with only one slot of carriers, paddle speedboats behave "well" without any sort of redistribution. For example with hephaistos 10 paddle speedboat do enough damage to kill a carrier (140 damage vs 140 life) and in fact in the slot of 2 carriers we have 1 death and 100% life. Notice in this case Am of speedboats has only one column. And notice if we approximate 13,6 with 13 then 10 speedboats should not be able to kill a carrier so 14 is the "appropriate

approximation".

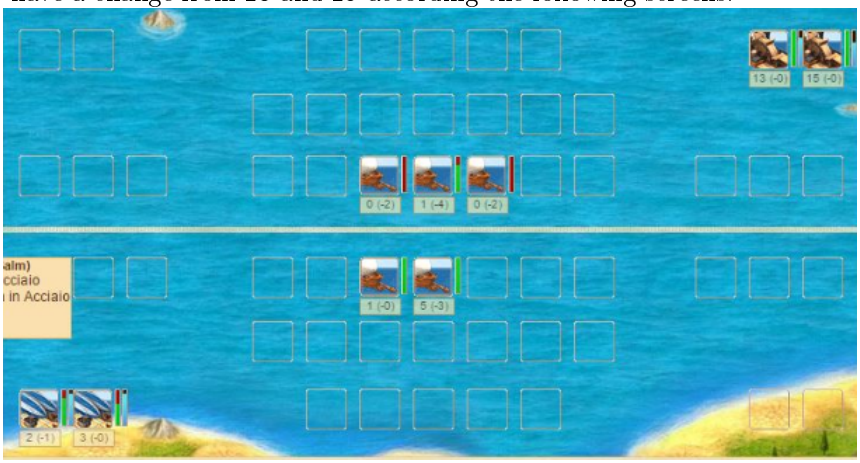
8.1 How to find redistribution patterns

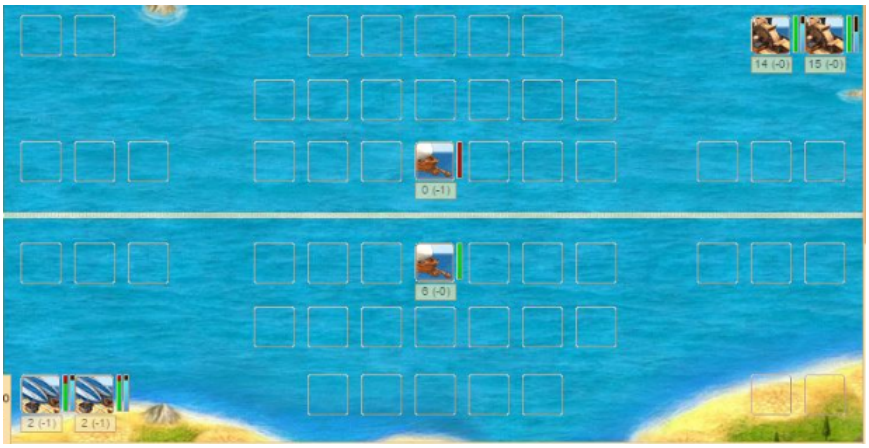
Suppose a line fires against another line. Some units perform IR, then other units hit sequentially an opponent's target until they kill it. This is not a kill whose definition I included in section 5, but we can sometimes hypothesize this behavior in paddle speedboats in this section, especially if we are not able to classify what type of kill happen (note we could simply throw all in M-kill or in IR-kill...).
So to find quickly a possible redistribution pattern you have to solve the following system:

$$\begin{cases} rx + ay \geq l \\ x + y \leq n \\ x \in \mathbb{N} \qquad y \in \mathbb{N} \end{cases}$$

where x is the number of units perform IR; y is the number of units hit sequentially; n is the total number of units hit in the slot; a is the attack value of a unit; r is the attack value of a unit redistributed on the slot, namely $r = \frac{a}{\text{units in the targeted slot}}$; l is the life of a units in the slot target.
Note if you want a "perfect" redistribution pattern you can solve the system with $=$ instead of $\geq$ in the first row.

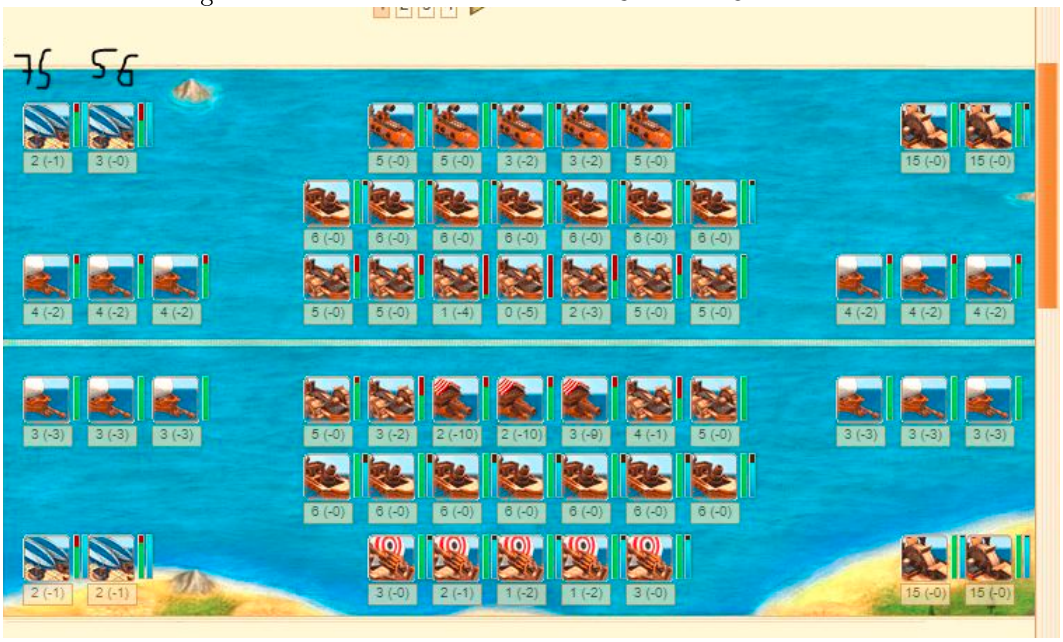
Now let's pass to the over gl field with 2 slots of 3 balloon carriers and 2 slots of paddle speedboats. If you have 30 speedboats in battle the corresponding attacking vector is [16,14]. If you have 29 the vector is [15,14] and if you have 28 the vector is [15,13]. Without hephaistos carriers never die, but with hephaistos we have a change from 28 and 29 according the following screens:





So in the second position of the vector, 14 speedboats do 196 total damage damage while 13 do 182 total damage. So we can suppose the first 6 speedboats redistribute damage(IR) in [5,5,4] and then 8 speedboats kill one carrier (30 redistributed damage + 112 > 140) but 7 speedboats are not able to kill (30 + 98 <140). As for damage interval in the 14 speedboats case it is [54,56] so we can not know if 2 of 142 damage is reassigned.

Now we have a screen with a level 4 hephaistos for the defender. In this case ↑ speedboats do 13 damage but the attacking vector is a normal [16, 14]. So the total damage on the second slot of carriers is $13 * 14 = 182$.



And with the previous schema of redistribution (first 6 speedboats redistribute [5,4,4]) we can again explain this situation in fact $30 + 13 * 8 = 134 < 140$.

Then consider a battle with no hephaistos: no carriers die and as life percentages we have [54%, 60%] . The total damage on slot 1 is $12 * 16 = 192$.

This time the previous redistribution schema fails (24+120>140). Moreover by resolving $4x + 12y = 140$ we have $x = 2$ and $y = 11$ as a integer solution (2 speedboats should redistribute and 11 not); so there would be a compatible redistribution scheme but the battle system in this case ignores it.

We can also consider the following situation: (wlog) attacker with hephaistos, defender with a slot of 15 hephaistosed speedboats and a slot of 15 non hephistofed speedboats. As for damage we have respectively 14 and 10. I do not include the screen, but no carriers die, and life percentages are [54%, 60%]. Even in this case total damage in carrier slot 1 is $192 = 14 * 8 + 10 * 8$

Now this screen:

where attacker is without hephaistos and his carriers have only +2 as defensive up, so a hephe speedboat inflict 20 damage. In both carriers slot no redistribution happens (note 140 is divisible by 20)
(This screen is also a direct proof that 2 rocket ships do not kill any steam rams because defender's slot 3 of steam rams have no losses).

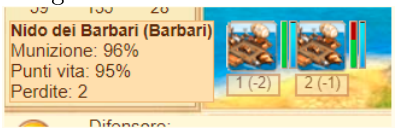
Finally consider the case of 5 carriers. In this case we have [60%, 40%] life percentage corresponding to an attacking vector [18,12]. Pay attention: now we have deduced a posteriori the attacking vector from life (and damage) percentages instead applying Am theory.
Then consider this screen:

Attacker without hephaistos, defender with hephaistos. A ↑ paddle speedboat inflicts 16 damage. The attacking ↑ vector is still [18,12]. In carriers slot 1 speedboats perform D-kill ($16 * 9 = 144 > 140$) while in carriers slot 2 we have again D-kill and 3 speedboats firing after the carrier deaths and causing 48 damage corresponding to 35%. Notice the total possible damage would be $12 * 16 = 192$.

8.2 A battle vs barbarians

Attacker with hephaistos vs barbarians; round 1: damage [48%,35%]
Damage of a speedboat against a barbarian balloon = 17; In the slot with 48% damage is $132 (= 17 * 16 - 140)$. because if we suppose 131 it would correspond

to 47%, (see par 3) . So no damage is lost. A possible redistribution scheme is 12 speedboats redistributed and 4 not $(12*6+17*4=140)$
In the slot with 35%damage interval is [96,98]. So at the end of the round total damage is between 228 and 230.
Barbarian fleet at level 49 has 36 carriers, and after first round there are 34 carriers still alive. So from RR, at the start of the second round carriers in the 2 slots have 20 or 21 damage. Now this is the second round.



5% damage in the left slot corresponds to damage interval [6,7], so at the start of the round we have a redistribution in each slot [7,7,z] where z can be 6 or 7. Then $17*8=136$ so in the left slot 2 carriers die without any damage recounted and no speedboat redistributes (D-kill).
In the right slot damage percentage is 48% corresponding to [118,120] as damage interval. In this case there are two possible behaviors: no redistribution and damage recounted (P-kill); or a redistribution scheme with 8 speedboats redistributing and 6 not. $(17x+6y=140-7=133 \longrightarrow x=5; y=8)$.

9 A rocket ship in the smallest battlefield (open problem)

Look at this 3 screens:

In the screen number 2 no steam rams die. This implies that damage of the rocket ship is redistributed (but we do not know if IR or SR) before steam rams fire. Moreover even damage of ↓ steam rams is redistributed but this time we can affirm with IR because otherwise total damage on the defender central slot would be enough to kill a steam ram.
Then in screen 1 a steam ram dies so the rocket or the attacking rams do not redistribution.
Note in screen 1 the defender's central slot is hit by 3 rams of the same column but different rows of the ↓ Am while in screen 2 the defender's central slot is

hit by 3 rams of the same row of the $\downarrow$ Am.
However we do not know if in screen 1 the situation is like in section 4.5 in the screen with archer and spearman where archer's damage is not redistributed; or if in this case rocket ship redistribute but steam rams no.
For a curious reader: if in the screen 2 you add a ballista ship you have a death in defender's steam rams. (and you have the same with a mortar ship, with a catapult ship or with a balloon carrier).

10 A swordsmen vs barbarian knifemen study (open problem)

10.1 Tests

We report results about the following battle-tests: we attacked an appropriate high level barbarian village in order to have an over gl battle; we sent swordsmen on the flanks and we observe the number of barbarian deaths on the flanks. For space-motivations we will include only two screens of the two types of tests : one-wave tests and more-wave tests.

10.2 Notation

First the Am of swordsmen is a 40x6 matrix. Imagine to isolate rows attacking on knifemen slot 1,2,3,4: you have 7 rows, while rows attacking slot 5 and 6 are only 6. So construct six reduced Ams, one for each knifemen slot with rows of the Am attacking on that slot.
A "strange" notation is chosen to represent zeros positions on the reduced matrices. Remember Ams and so reduced matrices have (according to this paper) only ones and zeros. We want to use a concise symbol to refer to different (reduced) Ams like in these examples.

$$\begin{bmatrix} 1 & \mathbf{0} & \mathbf{0} \\ 1 & 1 & \mathbf{0} \\ 1 & 1 & \mathbf{0} \end{bmatrix} = {}^{2}\urcorner_{3} \qquad \begin{bmatrix} 1 & 1 & 1 & \mathbf{0} & \mathbf{0} \\ 1 & 1 & 1 & 1 & \mathbf{0} \\ 1 & 1 & 1 & 1 & \mathbf{0} \end{bmatrix} = {}^{2}\urcorner_{3} \qquad \begin{bmatrix} 1 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 1 & 1 & 1 & \mathbf{0} \\ 1 & 1 & 1 & 1 \end{bmatrix} = {}^{3}\urcorner_{2}$$

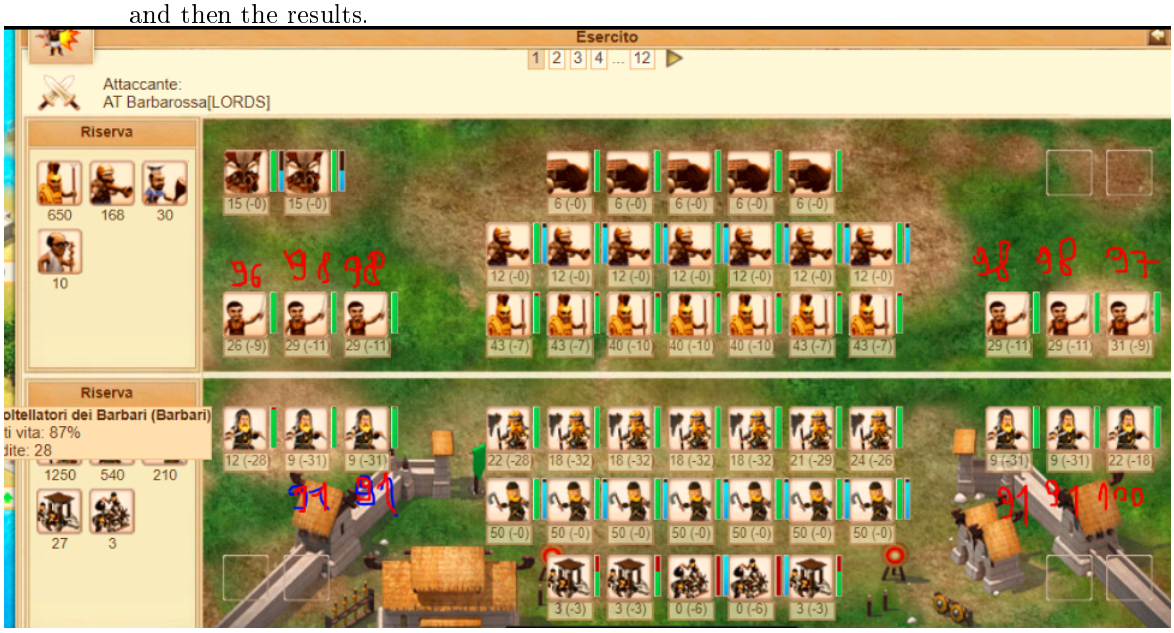
For example if you have only 200 swordsmen (5 slots of swordsmen) the sixth column of the Am is a zero column and so are sixth columns of reduced matrices.
Moreover we will also include 7 rows in reduced matrices, even when the seventh row is full of zeros.
This time we have chosen to start counting swordsmen from the low, so the eventually zeros row is the higher, but this is only a visual convention (actually this notation is in contradiction with Am theory notation and it is no-intuitive because the top Am row is written as the bottom in this notation for reduced matrices. I am sorry but these tests are have done before the writing of firsts sections of this paper).

The idea of these tests was trying to map different ways of kills according to different reduced Ams by a positional points of view: there can be reduced Attacking matrices with the same numbers of zeros but in different positions, so

at the beginning we suppose a whole reduced matrix have an impact on the kill method, not only the number of units it represents.

10.3 Tests with unique wave

In these tests swordsmen are sent all in an unique wave. We include a screen and then the results.



n. swordsmen (reduced matrices) → (killed barbarians$_{[\text{damage percentage}]}$)

201 $({}^{1}\urcorner_{6}, {}^{1}\urcorner_{7}, {}^{1}\urcorner_{7}, {}^{1}\urcorner_{7}, {}^{6}\urcorner_{7}, {}^{6}\urcorner_{7}) \rightarrow (31_{[12]}, 28_{[7]}, 28_{[7]}, 28_{[7]}, 15_{[0]}, 15_{[0]})$

205 $({}^{1}\urcorner_{6}, {}^{1}\urcorner_{6}, {}^{1}\urcorner_{6}, {}^{1}\urcorner_{6}, {}^{6}\urcorner_{6}, {}^{6}\urcorner_{7}) \rightarrow (31_{[12]}, 31_{[12]}, 31_{[12]}, 31_{[12]}, 26_{[12]}, 15_{[0]})$

207 $({}^{1}\urcorner_{5}, {}^{1}\urcorner_{6}, {}^{1}\urcorner_{6}, {}^{1}\urcorner_{6}, {}^{6}\urcorner_{6}, {}^{6}\urcorner_{6}) \rightarrow (29_{[8]}, 31_{[12]}, 31_{[12]}, 31_{[12]}, 26_{[12]}, 26_{[12]})$

209 $({}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{6}, {}^{6}\urcorner_{6}, {}^{6}\urcorner_{6}) \rightarrow (29_{[8]}, 29_{[8]}, 29_{[8]}, 31_{[12]}, 26_{[12]}, 26_{[12]})$

211 $({}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{6}\urcorner_{5}, {}^{6}\urcorner_{6}) \rightarrow (29_{[8]}, 29_{[8]}, 29_{[8]}, 29_{[8]}, 16_{[0]}, 26_{[12]})$

212 $({}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{6}\urcorner_{5}, {}^{6}\urcorner_{5}) \rightarrow (29_{[8]}, 29_{[8]}, 29_{[8]}, 29_{[8]}, 16_{[0]}, 16_{[0]})$

213 $({}^{1}\urcorner_{4}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{1}\urcorner_{5}, {}^{6}\urcorner_{5}, {}^{6}\urcorner_{5}) \rightarrow (32_{[12]}, 29_{[8]}, 29_{[8]}, 29_{[8]}, 16_{[0]}, 16_{[0]})$

217 $({}^{1}\urcorner_{4}, {}^{1}\urcorner_{4}, {}^{1}\urcorner_{4}, {}^{1}\urcorner_{4}, {}^{6}\urcorner_{4}, {}^{6}\urcorner_{5}) \rightarrow (32_{[12]}, 32_{[12]}, 32_{[12]}, 32_{[12]}, 27_{[12]}, 16_{[0]})$

219 $({}^{1}\urcorner_{3}, {}^{1}\urcorner_{4}, {}^{1}\urcorner_{4}, {}^{1}\urcorner_{4}, {}^{6}\urcorner_{4}, {}^{6}\urcorner_{4}) \rightarrow (30_{[8]}, 32_{[12]}, 32_{[12]}, 32_{[12]}, 27_{[12]}, 27_{[12]})$

223 $({}^{1}\urcorner_{3}, {}^{1}\urcorner_{3}, {}^{1}\urcorner_{3}, {}^{1}\urcorner_{3}, {}^{6}\urcorner_{3}, {}^{6}\urcorner_{4}) \rightarrow (30_{[8]}, 30_{[8]}, 30_{[8]}, 30_{[8]}, 17_{[0]}, 27_{[12]})$

229 $({}^{1}\urcorner_{2}, {}^{1}\urcorner_{2}, {}^{1}\urcorner_{2}, {}^{1}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{3}) \rightarrow (33_{[13]}, 33_{[13]}, 33_{[13]}, 33_{[13]}, 28_{[13]}, 17_{[0]})$

235 $({}^{1}\urcorner_{1}, {}^{1}\urcorner_{1}, {}^{1}\urcorner_{1}, {}^{1}\urcorner_{1}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{2}) \rightarrow (31_{[9]}, 31_{[9]}, 31_{[9]}, 31_{[9]}, 18_{[0]}, 28_{[13]})$

240 $({}^{0}\urcorner_{0}, {}^{0}\urcorner_{0}, {}^{0}\urcorner_{0}, {}^{0}\urcorner_{0}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{1}) \rightarrow (35_{[0]}, 35_{[0]}, 35_{[0]}, 35_{[0]}, 18_{[0]}, 18_{[0]})$

10.4 Explanations and observations

- In the battle cr life percentage is reported but I report damage percentage (100-life) because for a ease notation it is a smaller number. Results are showed in this way: on the right columns we have$(s_1, s_2, s_3, s_4, s_5, s_6)$ where s_i is the $i-th$ slot according to the standard renumbering. So this time visual correspondence with battle reports is not followed.
- Observing test 240 we can prove the "appropriate approximation" for swordsmen damage: barbarians have 0 damage so we can not have P-kill or M-kill. Obviously we can not have 1-kill nor D-kill because barbarians losses are 35. So the remaining possibilities are IR-kill or I-kill. If we approximate 15,6 with 16 the "possible redistribution" equation $2x+80y = 175$ has no integer solution; moreover the division $\frac{16*42}{18}$ is not integer. But if we approximate 15,6 with 15 the division $\frac{15*42}{18} = 35$ is integer.
- In slots where we have 0 damage we have D-kill (see section 4) except for test 240: in the first 4 slots we have I-kill and in the last 2 slots we have D-kill.
- From these test we can find a pattern: down index even $\rightarrow$ more losses; down index odd $\rightarrow$ less losses.

10.5 Tests with more waves

For these tests we needed 7 action points: first we send units without flanks to start the battle and then we sent 6 waves of swordsmen for the second round. We include a screen, waves and then results.

1. 36-36-36-36-36-36
2. 36-36-36-36-36-30
3. 37-36-36-36-36-30
4. 40-40-38-36-36-36
5. 40-39-38-37-36-35
6. 40-39-38-37-36-34
7. 40-38-38-36-36-33
8. 40-36-36-36-36-30
9. 39-36-36-36-36-30

1 $({}^{6}\urcorner_{1}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{1}) \rightarrow (18_{[0]}, 18_{[0]}, 18_{[0]}, 18_{[0]}, 18_{[0]}, 18_{[0]})$

2 $({}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}) \rightarrow (17_{[4]}, 17_{[4]}, 17_{[4]}, 17_{[4]}, 17_{[4]}, 17_{[4]})$

3 $({}^{5}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}) \rightarrow (29_{[7]}, 17_{[4]}, 17_{[4]}, 17_{[4]}, 17_{[4]}, 17_{[4]})$

4 $({}^{3}\urcorner_{1}, {}^{3}\urcorner_{1}, {}^{4}\urcorner_{1}, {}^{4}\urcorner_{1}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{1}) \rightarrow (32_{[5]}, 32_{[5]}, 21_{[6]}, 21_{[6]}, 18_{[0]}, 18_{[0]})$

5 $({}^{2}\urcorner_{1}, {}^{3}\urcorner_{1}, {}^{4}\urcorner_{1}, {}^{5}\urcorner_{1}, {}^{6}\urcorner_{1}, {}^{6}\urcorner_{2}) \rightarrow (33_{[4]}, 32_{[5]}, 31_{[5]}, 18_{[5]}, 18_{[0]}, 19_{[6]})$

6 $({}^{2}\urcorner_{1}, {}^{3}\urcorner_{1}, {}^{4}\urcorner_{1}, {}^{5}\urcorner_{1}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}) \rightarrow (33_{[4]}, 32_{[5]}, 31_{[5]}, 18_{[5]}, 19_{[6]}, 19_{[6]})$

7 $({}^{3}\urcorner_{1}, {}^{3}\urcorner_{1}, {}^{5}\urcorner_{1}, {}^{5}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}) \rightarrow (32_{[5]}, 32_{[5]}, 18_{[5]}, 20_{[6]}, 19_{[6]}, 19_{[6]})$

8 $({}^{5}\urcorner_{2}, {}^{5}\urcorner_{2}, {}^{5}\urcorner_{2}, {}^{5}\urcorner_{2}, {}^{6}\urcorner_{2}, {}^{6}\urcorner_{2}) \rightarrow (20_{[6]}, 20_{[6]}, 20_{[6]}, 20_{[6]}, 19_{[6]}, 19_{[6]})$

9 $({}^{5}\urcorner_2, {}^{5}\urcorner_2, {}^{5}\urcorner_2, {}^{6}\urcorner_2, {}^{6}\urcorner_2, {}^{6}\urcorner_2) \to (29_{[7]}, 29_{[7]}, 29_{[7]}, 17_{[4]}, 17_{[4]}, 17_{[4]})$

10.6 Observations

- Especially from tests 8,9 and 235 from tests with unique wave we see that swordsmen deaths depend not only on reduced matrices, despite our interpretation of reduced Ams in a positional sense.
  This was the first time when I understood the battle system is extremely hard to understand and I supposed losses and damage redistribution calculations depend on the whole battle field formation
- For the most interesting observation see next section (11)

10.7 An interesting round

Look at this screen:

In this battle barbarians attacked me at night but I had hephaistos active. Thanks to previous tests with one wave we know knifemen vs swordsmen have 0 damage, so at the start of round 10 of the screen barbarians flanks have 0 damage. Now we see a posteriori swordsmen are divided in 5 reduced Ams of 48 and perform D-kill. But we also know a swordsman do 15 damage and $15*48 = 720 = 18*40$ that means if swordsmen would perform I-kill they would be able to kill all knifemen. So why they perform D-kill?? We will continue discuss this screen in the next section.

We also observe 240 swordsmen vs 240 knifemen kill 176 units while 240 swordsmen vs 200 knifemen kill only 120 units.

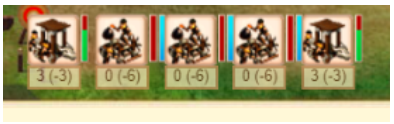
11 Columns role in Am (open problem)

First of all go back to the previous section and observe reduced matrices of swordsmen with the up index = 6. They are Ams with a row of zeros.Now we are interested in count the number of non-zeros in columns and it is easy to do the subtraction (length of a column)- (number of zeros=down index). It is also easy to notice that from the odd length=7 follows an even number of zeros corresponds to an odd number of non-zeros and vice versa.
So looking at out results we can affirm:
when all columns have an even number of non-zeros swordsmen perform D-kill (2 swordsmen in a column kill a knifeman with 0 remaining damage).
This situation happens also in the last screen with only 5 slots of knifemen where swordsmen columns have length=8.

Now look at this first round of a battle you can enterely find here
https://ikalogs.ru/report/e730802cc16e11eab4e9901b0e91886e/1/

Am of balloons of a column vector of length=15 and reduced matrices shooting in each rams slot are columns vector of length=5.
The attacker is with hephe so a balloon does 60 damage and life of a barbarian ram in 70 so is we would assume IR-kill we should have some rams killed (exactly 4). We deduce every balloon shoots on a different ram, or if you prefer, the system decides a priori no barbarian rams should die. Then from round 2 to 4 every balloon firing kills a ram and after the first round RR life percentages of rams is always the same (57%)

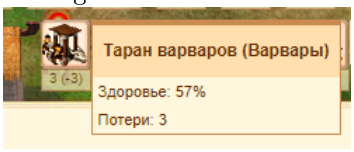
Continuing talking about balloons and rams we can consider an over GL battle without hephe where a balloon inflict 50 damage to an barbarian ram and there is no "appropriate approximation" issues:



where damage percentages of rams are 36% so the damage interval is [74,75] (means both 74 and 75 gives 36% damage, according to section 3). This time Am is a 15 x 2 matrix and ins vector notation is [6,6,6,6,6]. In each third line slot we have 2 columns of 3.
At this point we can explain rams deaths in 2 ways:

- First way is IR. IR happens after any shoot so a total of 150 damage is divided for 6 rams $\rightarrow$ 25 damage each ram; then when the 4th, 5th and 6th balloon fire we have 50 +25= 75 $>$ 70 so 3 rams are killed (IR-kill) and $25 * 3 = 75$ damage remains. Notice when the third balloon fire we have at most 17 + 50 damage $<$70.

- Second way is the following: suppose first column fire and each balloon hit a different ram. Then a redistribution happens and finally the second column fires and a balloon kills a ram. This is probably the correct interpretation of the round. In fact we can repeat the battle with hephe so a balloon does 60 damage and the first way fails because the second balloon should be able to kill a ram (10+60=70). But the second way works: 3 balloons fire hitting different rams, then redistribution give 30 damage for barbarian and finally the remaining 3 balloons fire and kill without damage recounting. So at the end of the round damage still on rams is $30 * 3$ corresponding to 43% damage –> 57% life.



Now imagine to have a cr against a level 43 barbarian village when barbarians have 4 third line slots only one with catapults. You have 2 slots of 10 balloons and hephaistos at level 5 and notice this losses with life percentages : $[2_{71\%}, 6(catapult), 3_{57\%}, 2_{71\%}]$ A balloon with hephaistos makes 60 damage on a ram (to be correct, suppose 60 as the "appropriate aproximation" of 60,2). The vector notation for Am of balloons is [4, 6, 6, 4]. So on the external slots we have 2 columns of 2 balloons. Again the second way works: 120 damage of the first column redistributes on 6 rams $\rightarrow$ 20 damage each, then the other column of 2 balloons kill 2 rams and $20 * 4 = 80$ remaining damage correspondent to 29% damage–> 71% life.

Now this is a second round of an hephaistos vs hephaistos battle.

Both player have 18 rams and in the first round, both have 5 balloons alive doing 275 damage. So at the start of the second round we have 16 or 15 damage on each ram (RR but the division is not integer).Then the vector notation for ↑ balloons Am is [3,2] but in this case Am and reduced Ams are column vectors. If we use the first way(IR-kill) 3 ↑ balloons should success in kill one ram because of 15 damage+ 2 balloons IR on the slot = 33 (at least) so 55+33=88 = life of a ram. But unfortunately 0 ram die. Note that with other approximations the final result is bigger than 88, so a ram should however die. So even in this case second way is the correct behavior. However continuing with rounds something change, in fact at the third round we have 550 damage redistributed on 18 rams → 30 or 31 damage. Then the vector notation for ↑ balloons Am is again [3,2].Again 31+55< 88 but at the end of the third round we have [2,1] as deaths of rams that can be explained this time by IR (first balloon redistribute, the other 2 kill). As for damage percentage at the end of round 3 we have [50%,47%] that means [[173,176], [203,206]] as damage intervals. And this means 43 or 44 damage on rams in the central slot but only 40, 41 or 42 damage on rams in external slot. And this is possible only with a P-kill (or PIR-kill) .

11.1 Tests with spearmen

To reproduce this type of tests you have to send one hoplite the first round, then enter in battle with a spearman before the end of the second round and retry the hoplite. Then enter in battle with waves of 1 spearman and wait until 4th round when life of barbarians will be 100% like in the first screen.

The 4th screen is from a different battle. Attacker with hephe so a spearman does 7 damage on a barbarian.
From these screens we see with a column of spearmen of length=4 there is no D-kill like in swordsmen. Same result with only a column:

At this point we can hypothesize a role of accuracy: in the last screen if accuracy of spearmen is 70% in the last screen 70% of 12 is 8 and in fact 8 spearmen kill 4 barbarians and the remaining 4 produce 28 damage –> 85% life.
Moreover balloons are very inaccurate while swordsmen are very accurate.

11.2 Paddle speedboats vs paddle speedboats

Now go back to the example of the RR formula in section 4.3.1, the battle between speedboats and barbarian speedboats. Our explanation of that situation fits with the "second way" of this section: there are 2 columns of attacker's speedboats. the first column is always able to use all 5 units to kill barbarians, so then the second columns fire independently. Anyway we know in a battle speedboats vs speedboats in the over GL battlefield the vector notation for Am is [14,16] so 2 columns of 8 fire in slot1 and 2 columns of 7 fire in slot 2. In slot 1 we always have 8 deaths and 100% , both with hephe or not, because 8 is even and each column performs D-kill; while in in slot 2 we have different behaviors with some remaining damage (M-kill).
An interesting test would be 20 speedboats vs 20 undamaged barbarians speedboats....

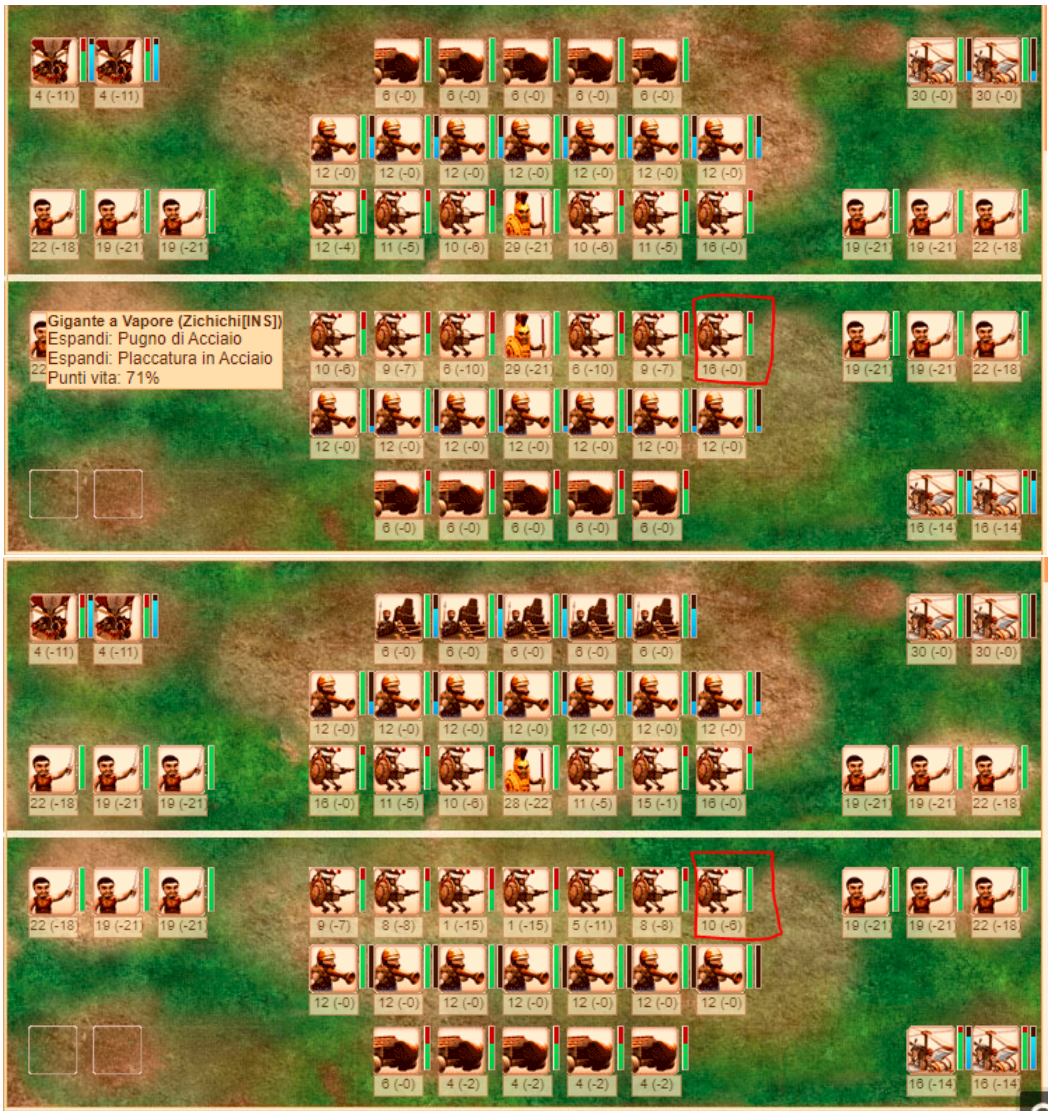
12 Beyond Am theory (very open problem)

12.1 Lunatic first line

Now we will confront two couples of screen. In the first couple look at the attacker's hoplites (**ignoring the red square in the first screen**):

It is the same battle except for the presence of steam giants in the second screen. But we well know first line is the last line firing, so ↑ hoplites firing on attacker's hoplites are exactly the same in both screens (apparently same Am, same reduced Ams). So why deaths in attacker's hoplites slot are different? In my opinion the key point is the diffrent slot composition when ↑ hoplites fire: a giant slot is complete while an hoplite slot is incomplete because of third line and second line previous fire.

Now the second couple: look at the giants in slot 7



We are sure giants in slot 7 are not injured at the start of the round in both screens. In the second screen the attacker have mortars but according to Am theory slot 7 of first line is not hit. So why slot 7 loses 6 giants?? In my opinion the key point is the different slot composition after ↓ third line fire: a mortar kill a giants while rams kill no giants.

13 Am theory lacks (very open problem)

In the following situations Am theory as described in this paper is not enough or is wrong:
- incomplete slots, for example in the following screen:
- weaker slots, for example spearmen in first line instead of hoplites or steam giants like in this following screen:

- total damage towards a slot is very bigger than total file of the slot and so same damage go to other slots. Often an incomplete slot or a weaker slot generates this situation, that we can see in the following screen in the defender:

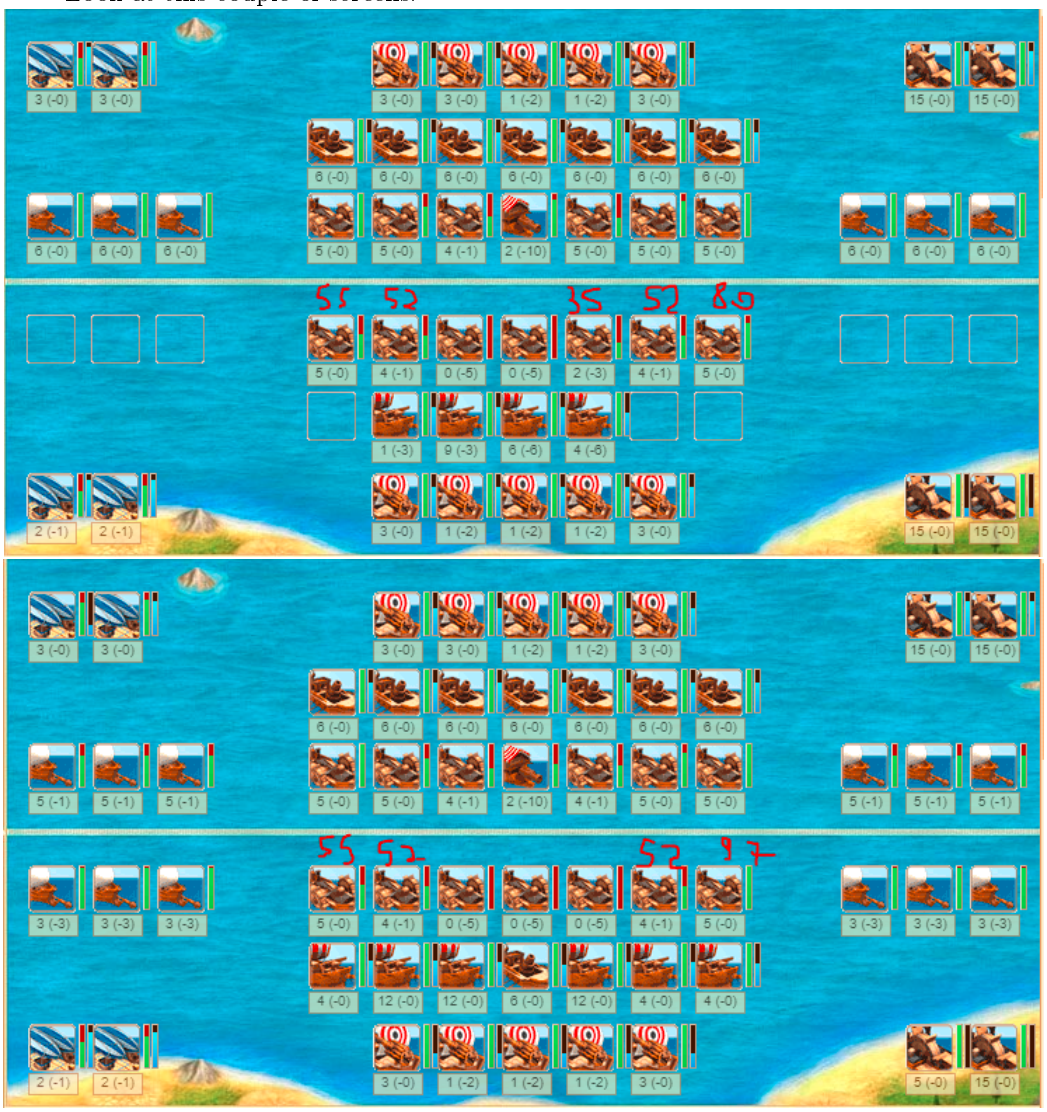
In a normal first line deployment, without incomplete slots we never have deaths in 6 slots (of course except you have 6 fire ships slots), but in this case the we have a slot with only 8 fire ships. They are probably not enough to "absorb" all damage that Am theory should give to that slot. It is not easy to talk about this argument: you can say the 8-fire ships slot is in position 7 of the standard renumbering, but clearly it took damage probably as slot 3. The difficulty is we cannot know a priori what renumbering is used. In this case incomplete slot not does not guarantee standard renumbering, and then there is also an other variation due to the presence of only 8 fire ships. You can try with 11 fire ships instead 8: there is also no standard renumbering but there are deaths only in five slots.

We also include this example in which we will focus on balloon carries and paddle speedboats:

The defender have enough tenders to be sure the 2 deployed carriers do not have any damage at the start of the round. Moreover attacker needs at least 20 speedboats to kill 2 carriers (280 life $= 14 * 20$). With a redistribution argument we can increase the number of speedboats firing on carriers of only one in fact if each speedboat redistributes its damage if possible we arrive at [133,133] damage and then we need 2 speedboats more to kill carriers. So speedboats firing on speedboats are 9 or 10.
Now we notice the defender speedboats loses and damage are symmetrical, but 9 is not divisible by 2. So suppose 10, that is divisible by 2, but apply Am theory. 10 means 5 rows; means 3 rows fire on slot 1 and 2 rows fire on slot 2 but this implies a non symmetrical behavior...
If we study losses and damage we see 6% corresponds to [14,15] so it seems 5 speedboats fire each defender's slot. So in this case Am theory fails.

13.1 The power of an absent second line (very open problem)

Look at this couple of screens:

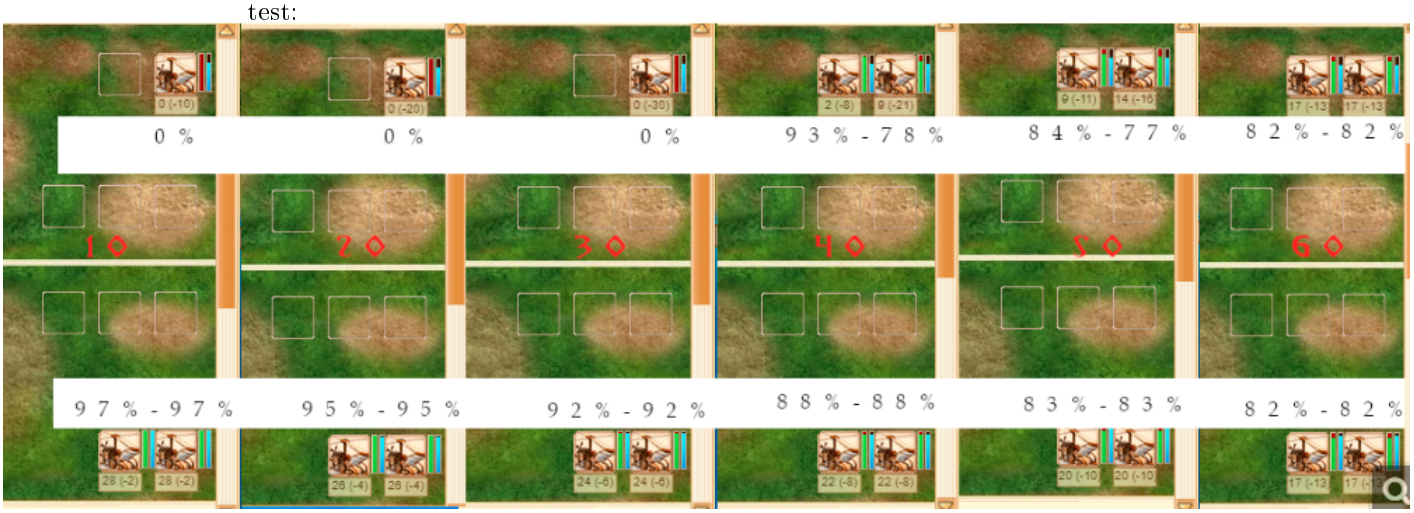


Attacker with hephe, defender without.
In the first screen attacker's flanks do not hit defenders first line because there are ballista ships still alive. So the attacker's formation firing against defender's steam rams is the same in both screen or is bigger in the first screen. But in the first screen the defender has less losses. However we observe in the first screen defender's steam rams slot 7 is more damaged than in the second screen so it seems in the first screen the attacker does not follow Am theory. So if this behavior is not a bug it seems that the result of a round is influenced by the whole battlefield deployment. Actually we do not know if is an issue of Am, reduced Ams, kill types or "damage target after slot 1 destruction".
We can repeat the same test with some variations but we always find less losses

in defender's steam rams when he does not have (a complete) second line:

14 Tests about gyrocopters (very open problems)

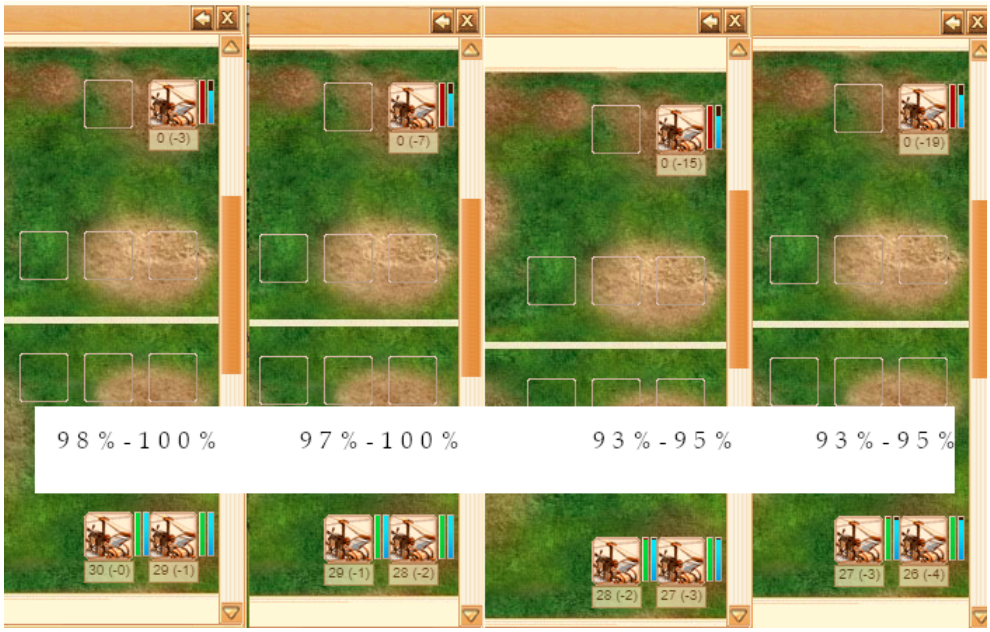
In this section we will show some gyrocopters-tests screens. Unfortunately we are not able to explain gyrocopters behavior. First screen is a no hephaistosed test:



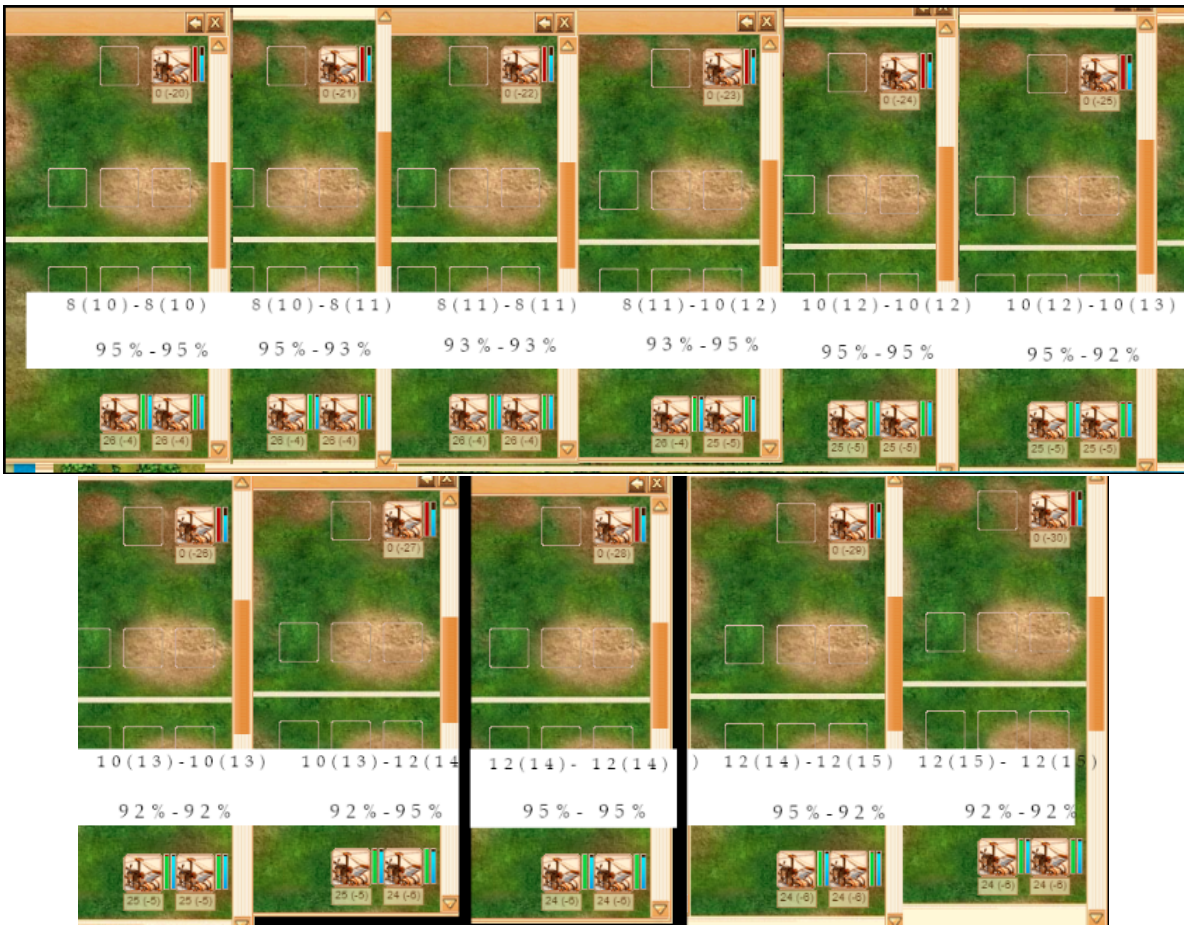
Then we report no hephaistosed tests with 60 gyrocopters vs from 3 to 30 gyrocopters. In the following table we have :
-N: total number of attacker's gyrocopters;
-Left: number of attacker's gyrocopters firing on the defender's left slot (according to AM theory);
-Right: number of attacker's gyrocopters firing on the defender's right slot (according to AM theory);
-mLeft: the minimum number of attacker's gyrocopters needed to produce losses we see in the defender's left slot (2 gyros kill one gyros);
-mRight: the minimum number of attacker's gyrocopters needed to produce losses we see in the defender's right slot (2 gyros kill one gyros);

-k-Left: number of gyrocopters killed in the defender's left slot;
-k-Right: number of gyrocopters killed in the defender's right slot;

N	$Left$	$Right$	$mLeft$	$mRight$	$k-Left$	$k-Right$
3	1	2	–	2	0	1
4	2	2	2	2	1	1
5	2	3	2	2	1	1
6	3	3	2	2	1	1
7	3	4	2	4	1	2
8	4	4	4	4	2	2
9	4	5	4	4	2	2
10	5	5	4	4	2	2
11	5	6	4	4	2	2
12	6	6	4	4	2	2
13	6	6	4	4	2	2
14	7	7	4	4	2	2
15	7	8	4	6	2	3
16	8	8	6	6	3	3
17	8	9	6	6	3	3
18	9	9	6	6	3	3
19	9	10	6	8	3	4

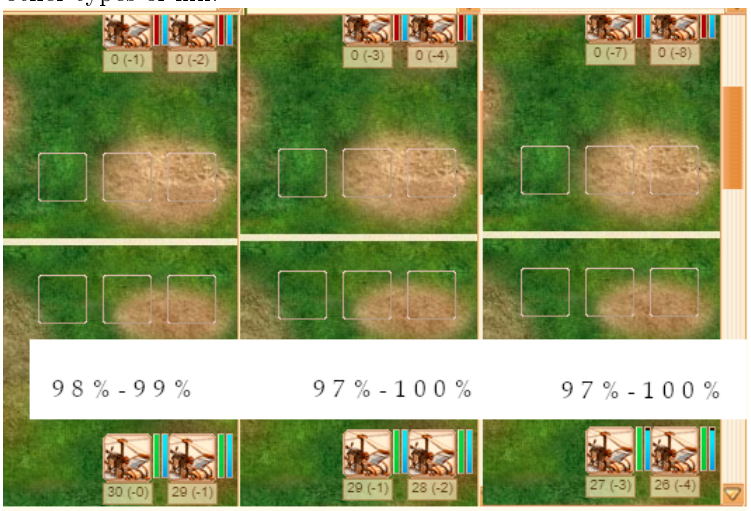


Then we include screens for test between 20 and 30 gyros:



All these tests present Am as a column so we can think about the 'second way' in section about columns role: each column perform D-kill, but with gyros we have an interesting exception: a column of length=6 kills 3 gyros and no damage left while a column of 8 gyros does not kill 4 gyros but only 3 and $17*2$ damage left. Then for longer columns every 2 more gyros kill one gyros, and for odd columns there also is one more gyros firing without killing. All percentages of the screens fit (but I do not report calculus).
Then we have also these tests with 2 columns:

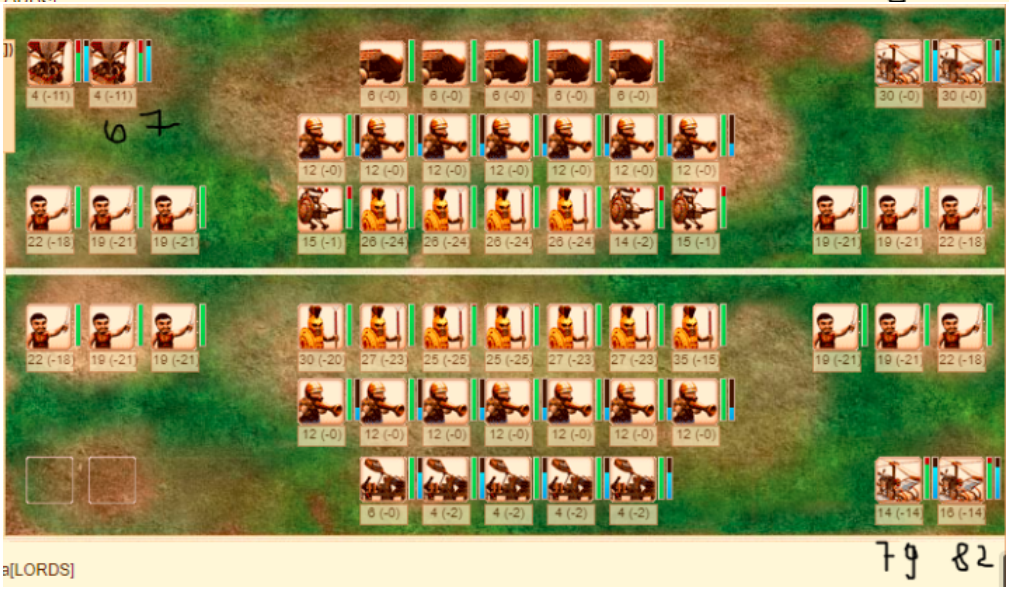
We could say first column fires, then some damage remain in the target slot and it could be redistributed (so we could introduce a new type of redistribution after a column fired); At this point the second column fires finding damaged units but not enough to1-kill them. So something more complicated than D-kill happen and we called it M-kill. In reality in our definition of M-kill it was thinked to explain situations like 60 gyros vs 60 gyros before the realization of a role of the column in the battle system. So it could be a combination of D-kill and an other types of kill.

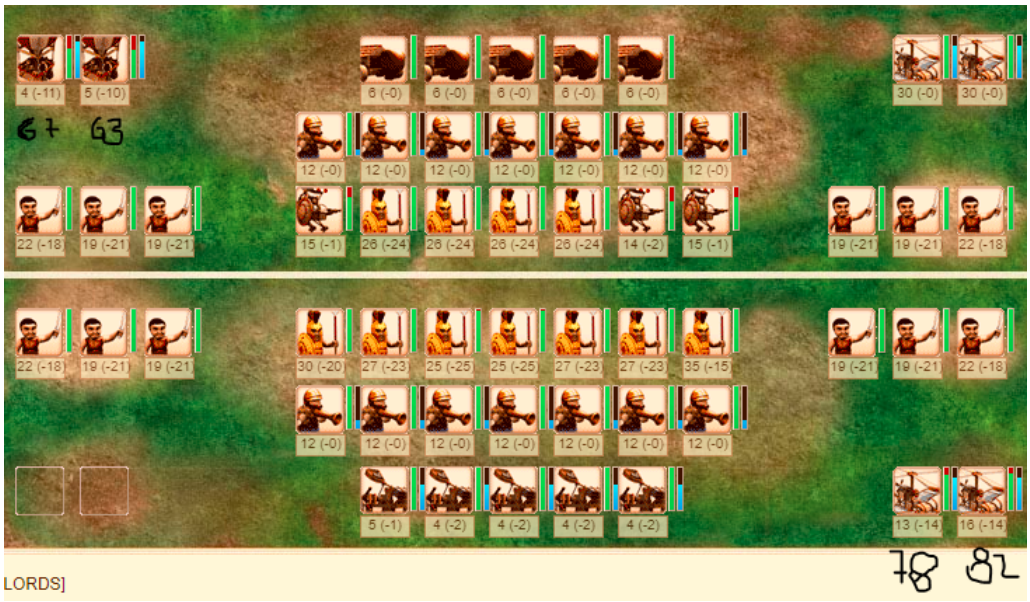


In this final screen on the left we have the only row of 2 gyros performs P-kill (note in this case there are 3 columns of length=1), while on the center and on the right we have D-kill by the columns and damage on the defender's gyros left slot caused by the odd column.

15 Symmetry issues in gyrocopters: do all units always fire? (very open problem)

We all know what happens in a battle with 60 gyros, but in this section we report 3 screens with 59, 58 and 57 gyros against balloons:

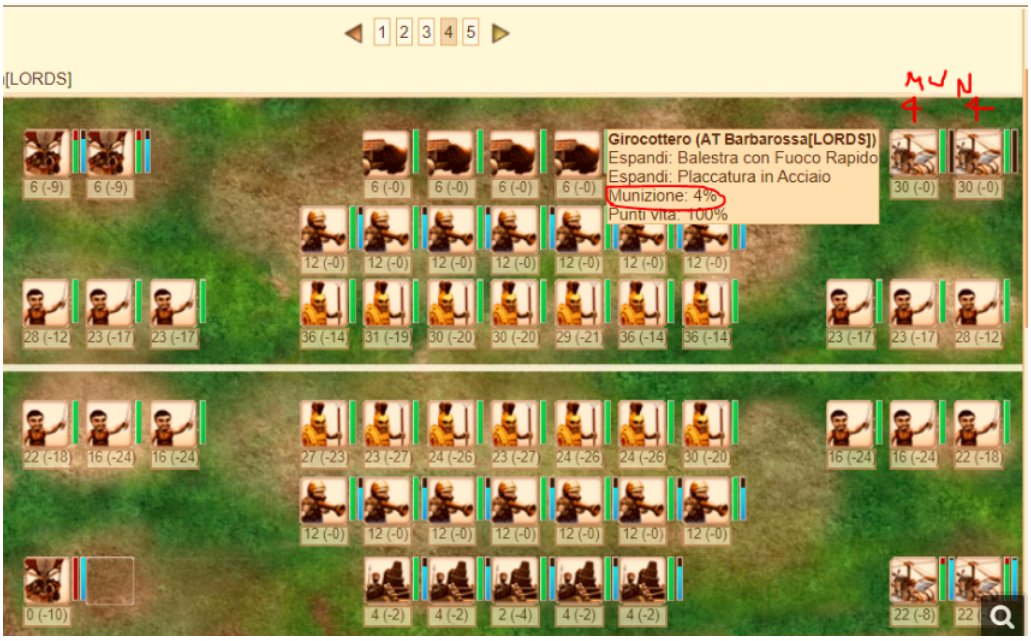




We also include a screen with 57 gyros vs gyros:

where the second round, on the right was for an evidence for redistribution. So why 60, 59 and 58 gyros inflicts same losses and damage against balloons? Why 57 gyros produce a symmetrical losses/damage against 60 gyros??

Now focus on this other screen that is the 4th round of a land battle: In the first 4 round attacker fires with the same 60 gyros, and their munitions are respectively 75, 50, 25 but then at the 4th round:



4% munition means 4 (or 5 because we do not know how approximate percentage of munitions) gyros per slot did not fire !!. This can explain speedboats behavior in the second screen of Am theory lacks section.

16 Two round of swordsmen without doctors (very open problem)

This is a 2 round battle. Attacker with hephaistos, defender without.

1% damage in the swordsmen central slots correspond to [1,2] damage interval. So the total damage on swordsmen after round 1 is 4 or 8. But swordsmen still

alive are 2596 so according our RR formula at the start of the second round each slot can have at max 1 damage.

And we see 1 damage cause a change in the attacker's swordsmen scheme with less deaths as a paradoxical result.
Note in an hephaistosed vs hepaistofed battle swordsmen without doctors increase losses from 21 to 22.
For future work: this was not a programmed test: repeat the same battle with third line in defender.

17 Duration tests (very open problem)

For duration tests I mean long battles with waves without doctors/tenders. For example gyros vs gyros, gyros vs balloons, spearmen without doctors vs spearmen and so on. You can find a lot of duration tests in https://ikalogs.ru/toplist/
For now we only report two example of duration test: first is gyros without hephaistos losses against gyros with hephe: N is the number of gyros in the wave at the start of the round; K_{left} are number of gyros losses in the left slot; L_{left} is the life on the left slot

Round	N	K_{left}	K_{right}	L_{left}	L_{right}
1	1282	15	15	[79	79]
2	1252	15	15	[75	75]
3	1222	16	16	[72	72]
4	1190	17	17	[77	77]
5	1156	18	18	[78	78]
6	1120	18	18	[77	77]
7	1084	17	17	[70	70]
8	1050	19	19	[77	77]
9	1012	19	19	[76	76]
10	974	18	18	[68	68]
11	938	19	19	[73	73]
12	900	20	20	[74	74]
13	860	20	20	[74	74]
14	820	19	19	[64	64]
15	782	22	22	[80	80]
16	738	22	22	[79	79]
17	694	22	22	[78	78]
18	650	21	21	[73	73]
19	608	22	22	[81	81]
20	564	23	23	[78	78]
21	518	23	23	[77	77]

The interesting point is the non-monotony of the K-columns (15,16,17,18,18,17,19,19,18...),(and consequently also of the L-column). Anyway calculations for these tests are the most difficult because even at the first round you start with a damage interval (section 3).

Another example of duration test is the following 8 round battle: 349 speramen vs 350 speramen:[6]
Round 1 / 8

Spearman.....276(-73) - Spearman.....278(-72)
Cook.....20(-0) - Cook.....20(-0)

Round2

[6] I did this test thinking about Lanchester's square law

Round 2 / 8

Spearman.....190(-86) - Spearman.....187(-91)
Cook.....20(-0) - Cook.....20(-0)

Round 3 / 8

Spearman.....128(-62) - Spearman.....122(-65)
Cook.....20(-0) - Cook.....20(-0)

Round 4 / 8

Spearman.....87(-41) - Spearman.....77(-45)
Cook.....20(-0) - Cook.....20(-0)

Round 5 / 8

Spearman.....60(-27) - Spearman.....48(-29)
Cook.....20(-0) - Cook.....20(-0)

Round 6 / 8

Spearman.....43(-17) - Spearman.....26(-22)
Cook.....20(-0) - Cook.....20(-0)

Round 7 / 8

Spearman.....31(-12) - Spearman.....6(-20)
Cook.....20(-0) - Cook.....20(-0)

Round 8 / 8

Spearman.....27(-4) - Spearman.....0(-6)
Cook.....20(-0) - Cook.....20(-0)

So the result is the victory of the weaker formation, and all start at round 2.
However to study duration tests we need a lot of tests and dates and actually I do not know how analyze these dates. Note RR formula produces no integer results so we should work with damage intervals and they would become bigger round after round.. a disaster. Moreover I think before duration tests we should study argument of section 11 and 12.

18 What about accuracy?

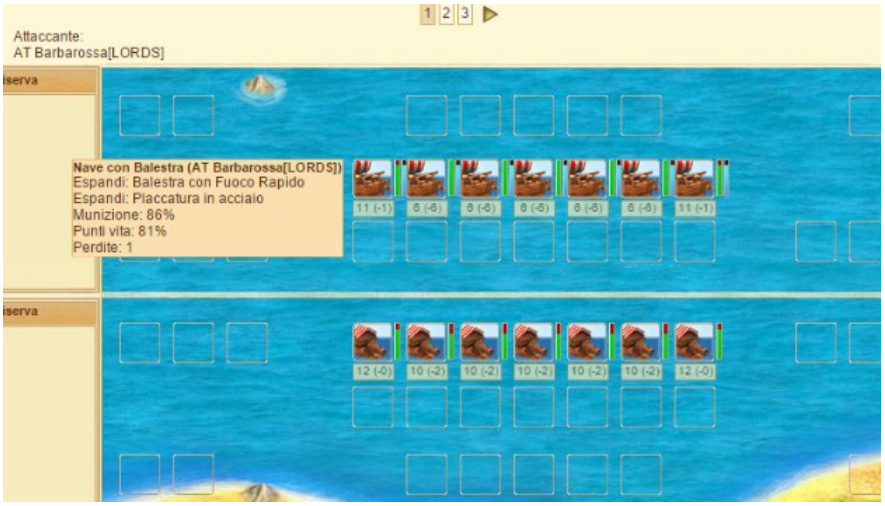
As you can see in this paper accuracy is almost never mentioned. In reality we have formulated and read a lot of accuracy-working hypothesis through years but for all there are some screens proving they are wrong. Anyway as you can see we were able to explain a lot of the Ikariam battle system ignoring accuracy so it does not seem an important thing. But it can be hide at an higher level. My last hypothesis were: -accuracy is related to IR happening; - accuracy in related to number of D-kill a column causes; - accuracy influences the Am and reduced Ams formation together with the battle field deployment (section of Am theory lacks).

19 Other questions

- How does Am and kill-types work when the Am should have different units in the same row? (for example both hoplites and steam giants)?
- How does target redistribution work when a first target slot is completely destroyed but according our Am theory there are other attacker's units should fire in that slot?
- How does target redistribution work when a first target line is completely destroyed and attacking units pass to another line?
- Does accuracy exist? How does it work?

20 Other useful facts in tests (bonus section)

- A tender cures 200 life points per round.
- Third line does not fire against second line. (see Ikariam in-game help)
- Second line fire with his white weapon only if it is attacked by first line. As an example you can look at this following screen with ballista ships: 14 ballista ships with only the distance-fire (ballista) would not do enough damage to kill 2 fire ships.



21 Authors and thanks

This paper was born by the will of AT Barbarossa of not waste years of screens, tests and reasoning about Ikariam battle system. And of course it was written by AT Barbarossa.
He started played Ikariam in 2009 in IT server ETA and now, after a fusion of server plays in IT server KAPPA.

First of all thanks to users of the old Italian forum, in particular Fly and Kain, whose posts about Ikariam battles I read during my first years of play.
Then many thanks to Xanto to support me with tests, theories and study so I included him as a co-author.
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